

AMS-SemiCD: A Synergistic Framework for Semi-Supervised Change Detection via an Adaptive Pseudo-Labeling Strategy and a Multi-Branch Network

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Abstract—Semi-supervised change detection (SSCD) aims to reduce the dependence on extensive labeled data. However, existing methods face three critical challenges. First, class imbalance and boundary ambiguity induce class-dependent confidence distributions, hindering reliable pseudo-label selection. Second, uncertain pixels may still contain useful structural cues, while their indiscriminate use can amplify pseudo-label noise. Third, achieving complementary local-global feature representations in dual-branch architectures typically incurs prohibitive computational costs. To address these issues, we propose the Adaptive Multi-branch Synergistic SSCD (AMS-SemiCD) framework, which uses Teacher–Student consistency regularization to supervise unlabeled bitemporal pairs. Within this framework, the Adaptive Statistical Pseudo-label Refinement (ASPR) strategy improves the reliability and utilization of consistency targets through class-adaptive dynamic thresholding and contribution-adaptive soft weighting. As a result, ASPR accommodates class-dependent confidence distributions while effectively exploiting uncertain but informative pixels. Meanwhile, the Multi-branch Synergistic Enhancement Network (MSE-Net) enhances discriminative feature representation by integrating local CNN and global Mamba branches with manageable computational costs. Specifically, MSE-Net consists of the Multi-branch Spatiotemporal Synergistic Fusion (MSSF) for local-global fusion, the Multi-branch Spatiotemporal Difference Booster (MSDB) for temporal-difference enhancement, and the Adaptive Gated Attention Refiner (AGAR) for cross-layer refinement. Extensive experiments on the GoogleGZ-CD, WHU-CD, and LEVIR-CD datasets demonstrate that AMS-SemiCD consistently outperforms advanced SSCD methods. Notably, with only 20% labeled data on LEVIR-CD, our method achieves an F1 score of 91.17% and an IoU of 83.78%. The code will be available at <https://github.com/RSII-NWNU/AMS-SemiCD>.

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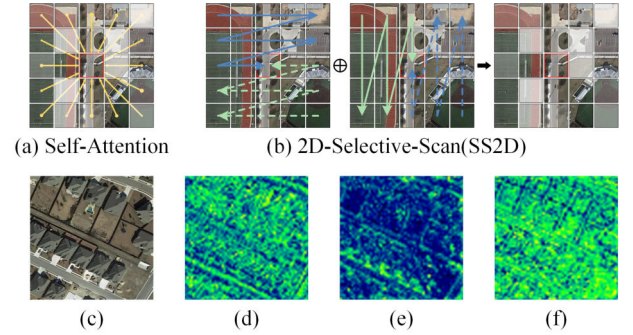


Fig. 1. Illustration of the computational mechanisms and feature representations. (a) Self-attention mechanism with quadratic $O(N^2)$ complexity. (b) 2D-Selective-Scan mechanism with linear $O(N)$ complexity. (c) Original input image. (d)-(f) Representative feature heatmaps visualizing the effectiveness of different branches. (d) Output of the VMamba branch, capturing global context but blurring boundaries. (e) Output of the CNN branch, preserving local details but missing long-range dependencies. (f) Output of the synergistic fusion, combining the strengths of both to form a robust representation.

Index Terms—Semi-supervised change detection, semi-supervised, change detection, remote sensing.

I. INTRODUCTION

Change detection is the process of identifying surface alterations by comparing remote sensing images (RSI) of the same area captured at different times. The change detection methodology is fundamental to a wide array of real-world applications, including urban planning [1], environmental monitoring [2], and disaster assessment [3].

In recent years, deep learning methods based on CNN [4] and Transformer architectures [5] have achieved significant success in CD tasks. However, improving the performance of these fully supervised change detection methods requires a large number of pixel-level annotated datasets. The creation of such labels is notoriously labor-intensive and costly, a dependency that severely hinders the practical deployment of supervised models in real-world scenarios. To mitigate this dependency, SSCD [6] has emerged as a promising research direction.

Currently, SSCD methods remain underdeveloped. Their primary objective is to efficiently extract latent change information from unlabeled image pairs. In general, semi-

supervised methods can be summarized into three main categories: self-training methods [7], adversarial learning-based methods [8], and consistency regularization-based methods [9]. Among them, consistency regularization enforces invariant predictions for unlabeled samples under randomized perturbations [10]. To obtain stable consistency targets, the Mean-Teacher framework [11] maintains an exponential moving average (EMA) Teacher network updated from the Student. The Teacher receives weakly augmented unlabeled samples and generates pseudo-labels. The resulting pseudo-labels are then used to supervise the Student under strong perturbations, forming the common consistency-based self-training paradigm. Since the effectiveness of this paradigm largely depends on pseudo-label reliability, recent SSL methods have introduced adaptive thresholding [12], [13] and confidence-based soft weighting [14] to improve pseudo-label selection and utilization. However, imbalance between changed and unchanged pixels and boundary ambiguity lead to class-dependent confidence distributions in SSCD. Representative strategies typically focus on either adaptive thresholding or soft weighting, while their joint regulation using class-wise confidence statistics remains underexplored.

Moreover, effectively exploiting uncertain pixels places higher demands on the feature extraction capability of the model. Single-branch architectures (pure CNN or Transformer) often struggle to balance local detail preservation with global context modeling. CNN excels at capturing fine-grained local details but is limited by their inductive biases in global modeling. Conversely, Transformer architectures capture long-range dependencies but can overlook crucial local details. Although hybrid CNN-Transformer architectures [15] seek to combine local and global advantages, the Transformer inherently relies on the self-attention mechanism. Self-attention computes dense token-to-token correlations, thereby introducing a prohibitive quadratic computational complexity $O(N^2)$, as illustrated in Fig. 1(a). This computational burden significantly restricts their applicability when processing high-resolution RSI. To overcome this bottleneck, recent advances have introduced Mamba, an architecture built upon State Space Model that utilizes the 2D-Selective-Scan (SS2D) mechanism as its core. Rather than relying on dense all-to-all computations, SS2D unfolds the 2D image into 1D sequences along four distinct directions, as visually compared in Fig. 1(b). It continuously compresses the historical context into a hidden state and selectively updates it based on the current input. This directional scanning and hidden state mechanism enables Mamba to establish global receptive fields with a linear computational complexity of $O(N)$. Consequently, the pivotal challenge lies in achieving the complementary local-global feature representation of dual-branch architectures with manageable computational costs.

To address these challenges synergistically, we propose the AMS-SemiCD framework, which leverages the Teacher-Student consistency regularization method as its core methodological basis. Our methodology tackles the semi-supervised bottleneck from both data supervision and architectural representation perspectives. From the data perspective, we introduce ASPR to track class-wise confidence statistics during training.

Class-adaptive dynamic thresholding establishes class-specific reliability criteria, while contribution-adaptive soft weighting assigns confidence-dependent contributions to uncertain pixels. A scheduled scaling factor coordinates both mechanisms, allowing broader utilization of uncertain predictions early in training and stricter reliability control later. From the architectural perspective, we design the Multi-branch Synergistic Enhancement Network (MSE-Net) to capture comprehensive spatiotemporal features. Unlike prior hybrid approaches that simply concatenate CNN and Transformer features [16] or use them as independent learners [17], MSE-Net achieves a layer-wise synergistic integration. Single branches exhibit inherent limitations, as evidenced by the feature visualization in Fig. 1(c-f). The Mamba branch (Fig. 1(d)) captures broad global context but blurs boundaries, while the CNN branch (Fig. 1(e)) precisely extracts local details but misses long-range dependencies. Leveraging Mamba's linear complexity (Fig. 1(b)), MSE-Net synergistically fuses these complementary features (Fig. 1(f)). The dual-branch design constructs robust local-global representations without the prohibitive computational costs inherent in Transformer-based hybrids.

The main contributions of this work are summarized as follows:

- 1) We introduce AMS-SemiCD, a synergistic framework combining MSE-Net for capturing complex bitemporal changes and ASPR for maximizing the utility of unlabeled data, thereby significantly improving SSCD performance.
- 2) We develop ASPR, which jointly derives class-adaptive dynamic thresholds and contribution-adaptive soft weights from class-wise confidence statistics, thereby adapting to class-dependent confidence distributions while effectively utilizing uncertain pixels.
- 3) We design MSE-Net to address single-branch feature extraction limitations. It synergistically integrates local and global features via MSSF, amplifies temporal differences via MSDB, and optimizes hierarchical reconstruction via AGAR.
- 4) Our proposed AMS-SemiCD achieves best results on three widely-used change detection datasets, outperforming seven competing SSCD methods. Notably, our method achieves performance comparable to that of fully supervised methods on the LEVIR-CD dataset, reaching an F1 score of 91.17% and an IoU of 83.78% while using only 20% of the labeled data.

The rest of this article is organized as follows. Section II presents the related work. In Section III, the proposed method is presented in detail. Section IV shows the experimental results. The conclusions are drawn in Section VI.

II. RELATED WORK

A. Deep Learning-based Change Detection

In the early stages, Daudt et al. [18] established a foundational baseline by proposing fully convolutional Siamese architectures to extract bitemporal features. To further optimize local feature representation, SFEARNet [19] combines semantic flow with edge-aware refinement. To compensate for

the inherent limitation of CNN in global context modeling, BIT [20] and ChangeFormer [5] introduced Transformer-based architectures. Both leverage self-attention mechanisms to effectively capture long-range spatiotemporal dependencies. For more robust representation, ConvFormer-CD [15] integrated CNN with Transformer for efficient temporal modeling, while SLAFormer [21] utilized skeleton-guided large-kernel attention to refine fine-grained details. Nevertheless, attention-based models often suffer from quadratic computational complexity. To overcome this bottleneck, ChangeMamba [22] pioneered the use of State Space Models, achieving efficient global context modeling with linear complexity. Building on this, Hybrid-MambaCD [23] further enhanced feature discriminability by designing an iterative global-local feature fusion strategy.

However, the heavy reliance on expensive annotations restricts practical deployment, highlighting the urgent need for semi-supervised approaches. Furthermore, distinguishing from prior Mamba-based works that often employ iterative or serial fusion mechanisms [23], our proposed MSE-Net adopts a parallel synergistic design. By leveraging Mamba's linear complexity to efficiently capture global context, our parallel design facilitates dense interaction with local CNN features. This synergistic architecture constructs robust local-global representations with manageable computational costs, which is critical for exploiting unlabeled data in semi-supervised settings.

B. Semi-supervised Learning

Semi-supervised learning (SSL) aims to improve model performance by leveraging a large amount of unlabeled data alongside a small set of labeled data [10]. A dominant paradigm in SSL is consistency regularization, which enforces that a model's predictions should remain stable under perturbations. Consistency regularization is often implemented through a student-teacher framework, where a teacher model provides stable supervisory signals for a student model [11]. To generate effective perturbations, RandAugment [24] performs structure-preserving augmentation on the input image, which improves representation robustness and generalization.

Self-training is another major paradigm of SSL, where model predictions on unlabeled data are used as training targets. FixMatch [25] combines self-training with consistency regularization and retains pseudo-labels above a fixed confidence threshold. To account for class-dependent learning progress, FlexMatch [12] introduces curriculum pseudo-labeling to dynamically adjust class-specific thresholds according to the learning status of each class. FreeMatch [13] further estimates global and class-specific learning states from unlabeled predictions to obtain self-adaptive thresholds. Beyond hard selection, SoftMatch [14] formulates pseudo-label utilization as sample weighting and applies a truncated Gaussian function to continuously weight predictions according to their confidence. Beyond confidence-based selection and weighting, recent studies have further improved pseudo-label reliability by reducing teacher-induced bias. For example, Na et al. [26] employ two differently initialized teachers to alleviate student overfitting to a single teacher.

FlexMatch [12] derives class-specific thresholds from curriculum learning progress, whereas FreeMatch [13] adapts a global threshold using unlabeled confidence and further modulates it according to class-wise learning states. Both methods retain threshold-based hard selection. In contrast, ASPR models the mean and dispersion of class-specific confidence distributions to jointly determine reliability thresholds and continuous contribution weights. SoftMatch [14] also adopts Gaussian weighting but parameterizes it using the overall confidence distribution. ASPR instead conditions Gaussian weighting on class-specific dynamic thresholds, with a scheduled scaling factor jointly controlling threshold adaptation and weighting decay.

Contrastive learning has also emerged as a powerful approach, which learns representations by pulling similar samples together and pushing dissimilar ones apart in the feature space [27]. Furthermore, adversarial learning using Generative Adversarial Network remains a relevant technique for generating realistic data augmentations and aligning feature distributions [28]. The successful application of these diverse SSL strategies has been demonstrated across various computer vision tasks, including semantic segmentation [29] and object detection [30].

C. Semi-supervised Change Detection

Change detection is a fundamental remote sensing task. Fully supervised deep learning methods have achieved remarkable accuracy [31], [32]. However, their reliance on large-scale annotated datasets remains a significant bottleneck. Consequently, SSCD has emerged as a crucial research direction to alleviate this annotation burden.

To compensate for the scarcity of supervisory signals, the field of semi-supervised change detection has evolved significantly. Initial efforts like SemiCDNet [8] preliminarily explored the efficacy of using network predictions to generate pseudo-labels. Subsequently, consistency regularization based on the student-teacher framework became the dominant paradigm. For instance, C2F-SemiCD [9] designed a coarse-to-fine consistency constraint strategy to enhance robustness in changed regions. UniMatch [33] further transferred weak-to-strong consistency regularization from semantic segmentation to the change detection task. Recently, LGTC [34] constructed a local-global tri-consistency network, reinforcing consistency constraints across different scales. To improve the effectiveness of perturbations, CutMix-CD [35] introduced the CutMix data augmentation strategy, compelling the model to focus on learning essential change-related features.

Since consistency learning relies on quality supervision, researchers have focused on enhancing the reliability of pseudo-labels. AdaSemiCD [36] employs an adaptive strategy based on pseudo-label evaluation to mitigate the accumulation of noisy labels. Similarly, Su et al. [37] introduced a boundary-refining teacher model to resolve boundary ambiguity within pseudo-labels. Concurrently, some scholars have also combined pseudo-labeling with contrastive learning to improve feature discrimination. For example, Guo et al. [38] proposed a confusion-aware contrastive loss to better distinguish ambiguous categories. Wang et al. [39] introduced a semi-supervised

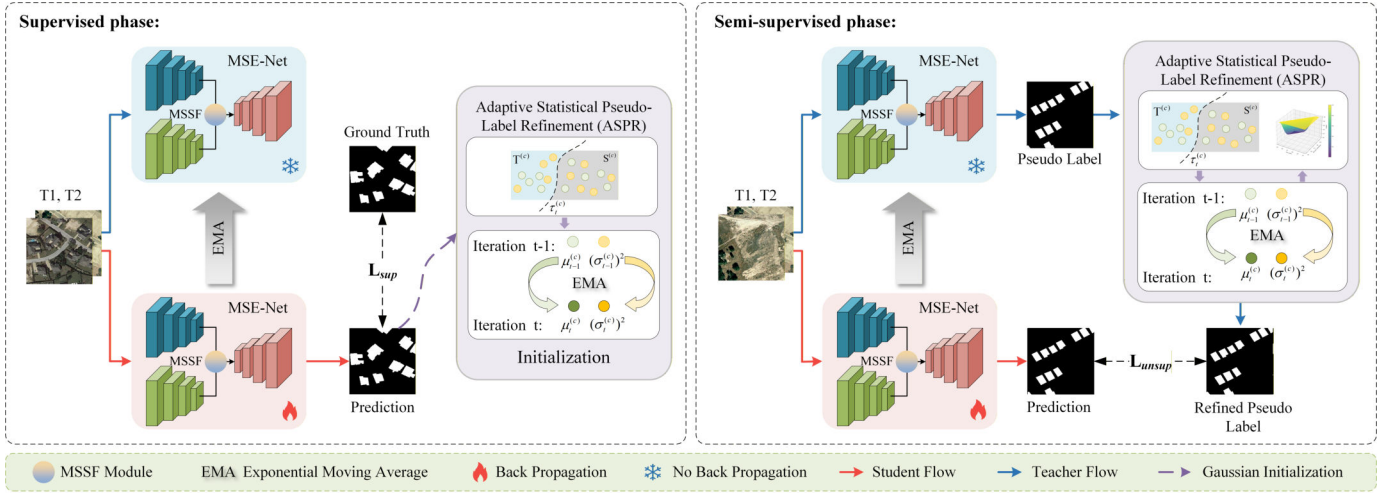


Fig. 2. Overall framework of the proposed AMS-SemiCD. The left side shows the training process during the supervision phase, where ASPR only initializes statistical parameters. The training process of the semi-supervised stage is shown on the right, where ASPR begins pseudo-label optimization.

domain alignment transformer that utilizes a dual-branch auto-encoder to align feature distributions despite limited labeled samples.

Moving beyond specific pseudo-label refinement strategies, the latest frontier in SSCD leverages powerful prior knowledge from large pre-trained foundation models. For instance, SemiCD-VL [40] improves the detection of complex changes by utilizing the vision-language model CLIP [41] to provide semantic guidance. Wang et al. [42] proposed a generative pretrained transformer for semi-supervised change detection that employs a spectral-spatial masked autoencoder to learn rich representations via self-supervised pre-training.

Despite these advances, pseudo-label refinement in SSCD still faces two task-specific challenges. In binary change detection, unchanged pixels dominate the training distribution, while changed pixels are sparse and boundary-sensitive, resulting in class-dependent confidence distributions. Existing SSCD approaches improve pseudo-label reliability from various perspectives, but the joint adaptation of selection thresholds and contribution weights to class-wise confidence statistics remains insufficiently explored. Moreover, uncertain pixels may contain useful boundary and structural cues that are lost under hard confidence filtering. Accordingly, ASPR jointly derives class-adaptive dynamic thresholds and contribution-adaptive soft weights from class-wise confidence statistics.

III. METHOD

In this section, we provide a detailed explanation of the proposed AMS-SemiCD architecture. The core of the framework consists of the ASPR strategy and the MSE-Net, which itself comprises the MSSF, MSDB, and AGAR modules.

Section III-A begins with an overview of the AMS-SemiCD method. Subsequently, Section III-B delves into the detailed design of ASPR. Section III-C elaborates on MSSF, Section III-D details MSDB, and Section III-E focuses on AGAR. Finally, Section III-F introduces the loss functions used in both the supervised and unsupervised stages, as well as the final composite loss.

A. Overall Architecture

AMS-SemiCD adopts the Mean-Teacher architecture for consistency regularization, featuring two parallel streams: a student and a teacher, as illustrated in Fig. 2. It functionally integrates two core innovations: a robust feature extractor (MSE-Net) and an adaptive supervision strategy (ASPR).

To capture discriminative changing features in complex RSI, we design MSE-Net as the shared architecture for student and teacher models, as shown in Fig. 3. Structurally, it comprises three key modules: MSSF integrates local CNN features and global Mamba dependencies in parallel to capture comprehensive context. MSDB enhances discriminative difference features to highlight change regions. AGAR bridges the semantic gap between layers for high-quality integration. Consequently, this architecture empowers the proposed framework to extract discriminative representations from both labeled and unlabeled data.

Complementing the feature extraction architecture, ASPR refines consistency targets through class-adaptive dynamic thresholding and contribution-adaptive soft weighting. It tracks class-wise confidence statistics to establish class-specific reliability criteria and assigns confidence-dependent contributions to uncertain predictions. A scheduled scaling factor coordinates the two mechanisms during training, allowing broader utilization of uncertain predictions at early stages and stricter reliability control as training progresses. The refined consistency targets are then used to supervise the Student under strong perturbations.

Finally, the overall training workflow proceeds in two distinct phases: a supervised phase and a semi-supervised phase, both involving coordinated student and teacher data flows. In the supervised phase, labeled image pairs are fed into the student network to calculate the supervised loss ($\mathcal{L}_{sup}$). The supervised training step enables the learning of basic change patterns and initializes statistical parameters for ASPR. Subsequently, in the semi-supervised phase, unlabeled image pairs are processed in parallel: weakly augmented views are fed into the teacher network to generate initial predictions,

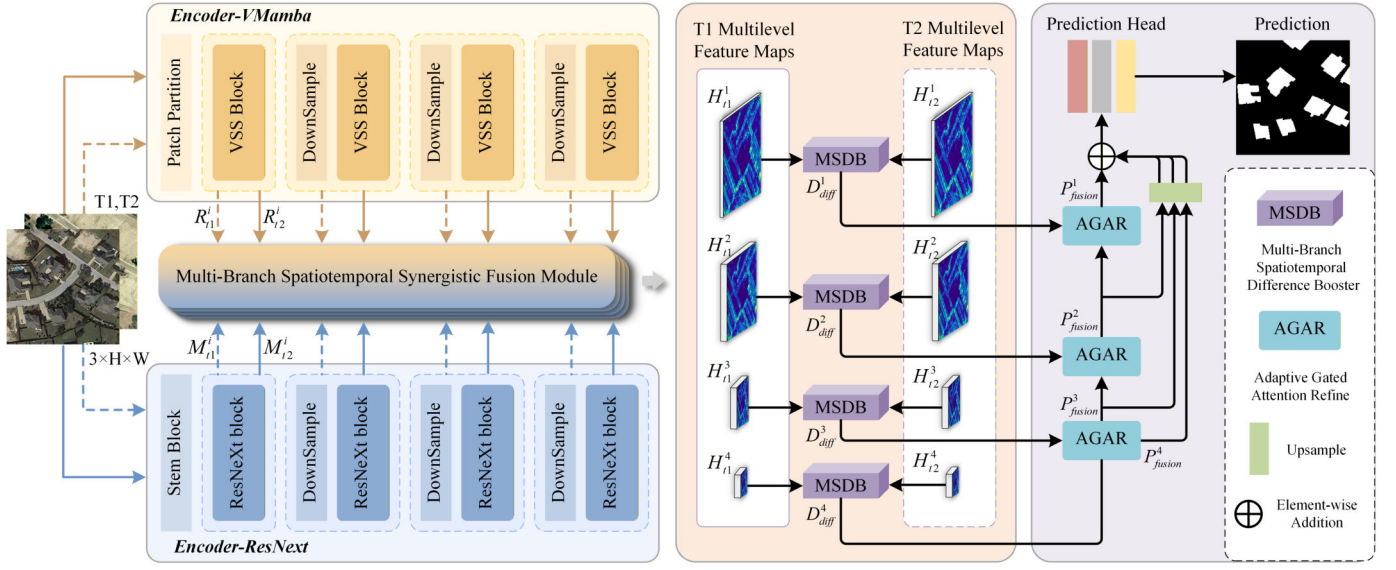


Fig. 3. MSE-Net network structure.

which are then refined by ASPR to remove noise and generate soft weights. Concurrently, strongly augmented views are fed into the student network. The refined pseudo-labels from the teacher network then supervise the student's predictions via the unsupervised consistency loss ($\mathcal{L}_{unsup}$). The synergistic design ensures that MSE-Net provides the capacity to learn, while ASPR provides the quality of supervision, collectively enabling efficient training on unlabeled data.

B. Adaptive Statistical Pseudo-Label Refinement (ASPR)

Class-dependent confidence distributions in SSCD complicate reliable pseudo-label utilization, while uncertain predictions may still contain useful information. To address these issues, ASPR computes and updates class-wise confidence statistics during training to jointly derive class-adaptive dynamic thresholds and contribution-adaptive soft weights. The former establishes class-specific reliability criteria, while the latter assigns confidence-dependent contributions to uncertain predictions during consistency learning.

Class-adaptive dynamic threshold. ASPR independently derives a confidence threshold for each class from the corresponding confidence statistics. The resulting threshold is continuously updated according to the class-wise confidence statistics to accommodate class-dependent confidence distributions during training. At the t -th iteration, the dynamic threshold for class c , denoted as $\tau_t^{(c)} (c \in \{\text{unchanged, changed}\})$, is calculated as follows:

$$\tau_t^{(c)} = \text{clamp} \left(\mu_t^{(c)} - \xi_t \cdot \sigma_t^{(c)} \right) \quad (1)$$

where clamp denotes the clamping operation, $\tau_t^{(c)}$ is clamped to the range $[0.5, 0.95]$. Theoretically, these boundaries establish a strict regularization constraint to form a safe trust region. Removing the 0.5 lower bound risks threshold collapse and subsequent gradient corruption by low-confidence noise [25]. Conversely, exceeding the 0.95 upper limit often induces

confirmation bias [43] and halts learning on difficult classes. This truncation effectively confines the optimization process within a stable functional space. ξ_t denotes the scaling factor, and t indicates the t -th iteration. $\mu_t^{(c)}$ and $\sigma_t^{(c)}$ represent the mean and standard deviation, respectively, of the predictions for class c . Based on the above calculation, we obtain a dynamic reliability benchmark $\tau_t^{(c)}$ that reflects both the confidence level and dispersion of class c .

$\tau_t^{(c)}$ is fixed at 0.5 to ensure stability during the first 5 rounds of training. Thereafter, $\tau_t^{(c)}$ is calculated according to Equation (1). The scaling factor ξ_t decreases according to a cosine schedule, yielding a more permissive reliability criterion early in training and a stricter criterion as training progresses. For the t -th iteration, ξ_t is calculated as follows:

$$\xi_t = \text{clamp} \left(1 + \cos \left(\pi \cdot \frac{E}{E_{\max}} \right) \right) \quad (2)$$

where ξ_t is clamped to the range $[0.5, 2]$, E represents the current training epoch, and $E_{\max}$ represents the total training epochs.

To estimate the statistical properties of the prediction distribution, ASPR designs an elegant procedure for constructing sample sets and updating statistical parameters. First, it attempts to construct a high-confidence prediction set $\mathcal{H}^{(c)}$ for class c . If the prediction size of $\mathcal{H}^{(c)}$ is insufficient, the method proceeds to construct an ordinary prediction set $\mathcal{S}^{(c)}$ with more lenient criteria. Based on this, re-estimate the distribution parameters $\mu_t^{(c)}$ and $(\sigma_t^{(c)})^2$ to approximate the true distribution. The prediction set construction at the t -th iteration is defined as

$$\begin{aligned}
\hat{y}_i &= \operatorname{argmax}_c \mathbf{p}_i^{(c)} \\
\mathcal{H}^{(c)} &= \left\{ i \mid \mathbf{p}_i^{(c)} > \tau_{t-1}^{(c)} \wedge \hat{y}_i = c \right\} \\
\mathcal{S}^{(c)} &= \left\{ i \mid \mathbf{p}_i^{(c)} > 0.5 \wedge \hat{y}_i = c \right\} \\
\mathcal{M}^{(c)} &= \begin{cases} \mathcal{H}^{(c)}, & \text{if } |\mathcal{H}^{(c)}| > 1, \\ \mathcal{S}^{(c)}, & \text{otherwise.} \end{cases}
\end{aligned} \tag{3}$$

where $\mathbf{p}_i^{(c)}$ denotes the predicted confidence of the i -th prediction for class c , $\hat{y}_i$ represents the predicted class label of the i -th prediction, and $\mathcal{M}^{(c)}$ represents the prediction set for class c . At the t -th iteration, the relevant distribution parameters are calculated as follows:

$$\begin{aligned}
\hat{\mu}_t^{(c)} &= \frac{1}{|\mathcal{M}^{(c)}|} \sum_{i \in \mathcal{M}^{(c)}} \mathbf{p}_i^{(c)} \\
(\hat{\sigma}_t^{(c)})^2 &= \frac{1}{|\mathcal{M}^{(c)}| - 1} \sum_{i \in \mathcal{M}^{(c)}} \left(\mathbf{p}_i^{(c)} - \hat{\mu}_t^{(c)} \right)^2
\end{aligned} \tag{4}$$

where $\hat{\mu}_t^{(c)}$ and $(\hat{\sigma}_t^{(c)})^2$ represent the estimated mean and variance of the prediction results for class c during iteration t . $\hat{\mu}_t^{(c)}$ characterizes the confidence level of the predictions for each class, while the $(\hat{\sigma}_t^{(c)})^2$ regulates the stringency of the selection criteria for the predictions of each class. It is worth noting that when $\mathcal{M}^{(c)}$ is composed of $\mathcal{S}^{(c)}$, the contribution of updating the class-wise statistics by $\mathcal{M}^{(c)}$ will be moderately attenuated. In semi-supervised learning, pseudo-labels derived from individual mini-batches often exhibit severe statistical variance. We theoretically employ the EMA to temporally smooth these stochastic batch-level fluctuations [11]. This mechanism suppresses abrupt deviations in the class-wise statistical estimates and stabilizes their updates. The update process is as follows:

$$\begin{aligned}
\beta &= \begin{cases} 1, & \text{if } |\mathcal{H}^{(c)}| > 1, \\ 0.1, & \text{otherwise.} \end{cases} \\
\mu_t^{(c)} &= m \cdot \mu_{t-1}^{(c)} + \beta \cdot (1 - m) \cdot \hat{\mu}_t^{(c)} \\
(\sigma_t^{(c)})^2 &= m \cdot (\sigma_{t-1}^{(c)})^2 + \beta \cdot (1 - m) \cdot (\hat{\sigma}_t^{(c)})^2
\end{aligned} \tag{5}$$

where m denotes the momentum parameter for the EMA update, and β represents the attenuation coefficient updated by EMA.

Contribution-adaptive soft weighting. After establishing the dynamic reliability benchmark $\tau_t^{(c)}$ for each class via the class-adaptive dynamic threshold mechanism, we do not simply discard predictions with confidence scores below this benchmark. We argue that those predictions with confidence scores near the threshold also contain substantial latent information worthy of further exploration.

To effectively leverage such latent information, we propose a contribution-adaptive soft weighting mechanism. Departing from traditional hard thresholding that categorically accepts or rejects pseudo-labels, this mechanism dynamically assigns a contribution weight to each pixel prediction in the loss function based on its prediction confidence. Specifically, the mechanism employs a Gaussian-based function conditioned on the

dynamic threshold to model the non-linear decay of prediction reliability. Compared with SoftMatch [14], ASPR conditions Gaussian weighting on class-specific dynamic thresholds, with the scheduled scaling factor ξ_t jointly regulating threshold adaptation and weighting decay. Compared to linear or step functions, the Gaussian kernel provides a smooth and differentiable penalty. It assigns higher relative weights to informative predictions near the threshold while exponentially suppressing low-confidence noise. This design avoids abrupt changes in prediction contributions caused by hard thresholding. The soft weighting mechanism preserves potentially useful information from predictions with confidences slightly below the threshold, thereby enhancing the stability of the learning process. Accordingly, the weighting function ω_i is formulated as follows:

$$\begin{aligned}
\hat{\mathbf{p}}_i &= \max_c \mathbf{p}_i^{(c)} \\
\Delta_i &= \left(\tau_t^{(\hat{y}_i)} - \hat{\mathbf{p}}_i \right)^2 \\
\omega_i &= \begin{cases} \exp \left(-\frac{\Delta_i}{2(\sigma_t^{(\hat{y}_i)})^2 \cdot \xi_t^2} \right), & \hat{\mathbf{p}}_i < \tau_t^{(\hat{y}_i)} \\ 1, & \hat{\mathbf{p}}_i \geq \tau_t^{(\hat{y}_i)} \end{cases}
\end{aligned} \tag{6}$$

where $\hat{\mathbf{p}}_i$ denotes the maximum prediction probability for the i -th pixel prediction. The soft weight ω_i is computed using the class-specific statistical parameters $\tau_t^{(\hat{y}_i)}$ and $\sigma_t^{(\hat{y}_i)}$ associated with the predicted class $\hat{y}_i$. Predictions with confidence scores exceeding $\tau_t^{(\hat{y}_i)}$ are assigned a weight of 1, whereas those below the threshold receive smoothly decaying non-zero weights rather than being directly discarded.

In summary, ASPR uses class-wise confidence statistics to couple class-adaptive dynamic thresholding with contribution-adaptive soft weighting. This formulation adapts pseudo-label utilization to class-dependent confidence distributions while retaining informative uncertain pixels. The overall procedure of ASPR is outlined in Algorithm 1.

C. Multi-Branch Spatiotemporal Synergistic Fusion (MSSF)

The task of change detection in high-resolution RSI relies on the accurate capture of two complementary features: local texture features and global semantic features. The effective fusion of the two is key to improving change detection accuracy. While current CNN-Transformer hybrid architectures attempt to integrate both, they are often limited by the quadratic computational complexity of Transformers. In contrast, our proposed MSSF module introduces a novel parallel architecture combining CNN and Mamba. The CNN branch captures local details, while the Mamba branch models long-range dependencies with linear spatial complexity, enabling complementary local-global feature integration.

The MSSF consists of a bidirectional channel-spatial attention and the exclusive feature injection (EFI), enabling fine-grained modeling of both common and specific features, as illustrated in Fig. 4. First, the features R_b^i and M_b^i ($i \in \{1, 2, 3, 4\}$, $b \in \{T1, T2\}$) from the CNN and Mamba branches

Algorithm 1 Adaptive Statistical Pseudo-Label Refinement

Input: Prediction set: $p_i^{(c)}$
 Number of classes: C
 Momentum coefficient: m (default = 0.999)
 Total training epochs: E_{max}
 Batches per epoch: $N_{batches}$
for $E \leftarrow 1$ **to** E_{max} **do**
for $b \leftarrow 1$ **to** $N_{batches}$ **do**
 $t \leftarrow (E - 1) \times N_{batches} + b$
 Compute scaling factor ξ_t using Equation (2)
for $c \leftarrow 1$ **to** C **do**
if $E \leq 5$ **then**
 Set selection threshold $\tau_{t-1}^{(c)} \leftarrow 0.5$
else
 Compute selection threshold $\tau_{t-1}^{(c)}$ using Equation (1) with $\mu_{t-1}^{(c)}, \sigma_{t-1}^{(c)}$
end if
 Construct prediction set $\mathcal{M}^{(c)}$ using Equation (3) with $\tau_{t-1}^{(c)}$
 Compute $\hat{\mu}_t^{(c)}$ and $\hat{\sigma}_t^{(c)}$ for the current sample set using Equation (4)
 Update $\mu_t^{(c)}$ and $\sigma_t^{(c)}$ via EMA using Equation (5)
end for
 Compute dynamic threshold $\tau_t^{(c)}$ for all classes using Equation (1) with $\mu_t^{(c)}, \sigma_t^{(c)}$
 Compute the soft weights ω_i using Equation (6)
 Refine pseudo-labels using ω_i
end for
end for
Output: refined pseudo-labels

undergo channel alignment to obtain $\bar{R}_b^i$ and $\bar{M}_b^i$. The process is mathematically formulated as

$$\begin{aligned}\bar{R}_b^i &= \text{GN}(\text{Conv1}(R_b^i)) \\ \bar{M}_b^i &= \text{GN}(\text{Conv1}(M_b^i))\end{aligned}\quad (7)$$

where GN denotes Group Normalization, and Conv1 represents a 1×1 convolution.

The common features are extracted jointly through bidirectional channel-spatial attention, with the overall structure illustrated in Fig. 4. In the channel dimension, the bidirectional cross-channel attention uses the CNN and Mamba branches as the respective leading sources to generate channel attention weights, thereby extracting bidirectional channel commonality features CR^i and CM^i . Similarly, in the spatial dimension, the bidirectional cross-spatial attention employs both branches to produce spatial attention weights, yielding bidirectional spatial commonality features SR^i and SM^i . Through mutual guidance, the bidirectional channel-spatial attention leverages the CNN and Mamba branches to capture common features. It enables each branch to prioritize the informative channels and spatial regions jointly emphasized by both modalities. Subsequently, SR^i and SM^i are summed element-wise, and the result is concatenated with CR^i and CM^i along the channel dimension. The fusion process further integrates complementary common information from both spatial and

channel dimensions. The combined features are then processed sequentially by a 3×3 depthwise convolution (DwConv3), a ReLU activation function, and a 1×1 convolution (Conv1) to obtain the common features F^i . The process can be formulated as follows:

$$F^i = \text{Conv1}(\text{ReLU}(\text{DwConv3}(\text{Concat}[CM^i, CR^i, SM^i \oplus SR^i]))) \quad (8)$$

where Concat signifies the concatenation operation, while the operators $\oplus$ represent element-wise addition.

The EFI integrates branch-specific residual information through an input pathway, efficiently extracts the exclusive features from each branch, as shown at the bottom of Fig. 4. First, EFI feeds the results of max pooling (MaxPool) and average pooling (AvgPool) into the Multilayer Perceptron (MLP), generating channel-wise weights via the Sigmoid function. The weights are applied to enhance the most significant channels in branch-specific features. Next, the Coordinate Attention (CA) [44] adopts an innovative direction-aware decoupled pooling strategy to enhance spatially relevant salient features. Since the exclusive features often contain noise and artifacts, we design a lightweight gating module composed of BaseConv, 3×3 convolutional (Conv3), and the Sigmoid function. BaseConv is composed of Conv1, batch normalization (BN), and ReLU. The gating mechanism effectively filters out noise interference, yielding denoised exclusive features GR^i and GM^i . The extraction of Mamba exclusive features can be formulated as follows:

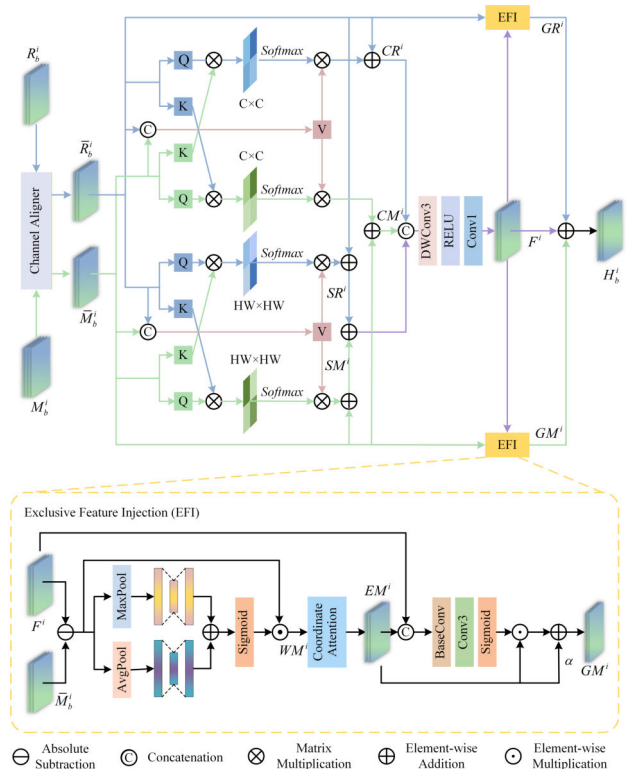


Fig. 4. Details of the MSSF module. The internal structure of the EFI block is illustrated at the bottom.

$$\begin{aligned}
WM^i &= \delta(\text{FC}(\text{MaxPool}(|\bar{M}_b^i - F^i|)) \\
&\quad \oplus \text{FC}(\text{AvgPool}(|\bar{M}_b^i - F^i|))) \odot |\bar{M}_b^i - F^i| \\
EM^i &= \text{CA}(WM^i) \\
GM^i &= \delta(\text{Conv3}(\text{BaseConv3}(\text{Concat}[EM^i, F^i]))) \\
&\quad \odot EM^i \oplus (EM^i \odot \alpha)
\end{aligned} \quad (9)$$

where δ denotes the Sigmoid activation function, $|\cdot|$ signifies the absolute difference operation, the operators $\odot$ represent element-wise multiplication, and α is the residual coefficient. WR^i and WM^i represent the intermediate features, and ER^i and EM^i correspond to the exclusive features at the i -th layer.

Furthermore, the EFI introduces adversarial gradient interference to the extraction of common features during back-propagation. The adversarial gradient interference arises from the conflicting objectives of the common and exclusive feature pathways. The common feature extractor aims to minimize the difference between branch features and the common representation, while the EFI seeks to maximize meaningful components of this same difference to highlight exclusive information. The resulting conflict creates contradictory gradient signals during backpropagation, destabilizing the optimization of shared features. The introduction of a learned damping coefficient helps to dynamically balance gradient flow, thereby suppressing harmful interference while preserving beneficial interactions.

The final fused feature H_b^i is obtained by element-wise addition of GR^i , GM^i and F^i . It is formulated as follows:

$$H_b^i = F^i \oplus GR^i \oplus GM^i \quad (10)$$

Based on this, the MSSF effectively integrates local textures with global semantics, achieving a synergistic complementarity between CNN and Mamba.

D. Multi-Branch Spatiotemporal Difference Booster (MSDB)

Conventional methods for generating difference maps from bitemporal images, such as the difference and concatenation approaches, suffer from inherent limitations. The difference method is highly sensitive to noise, often leading to the loss of fine-grained change details. The concatenation method achieves performance gains at the cost of reduced interpretability and increased data dependency [31]. Furthermore, registration errors arising from discrepancies in imaging viewpoints further undermine the reliability of the results. To address these issues, we propose a novel Multi-Branch Spatiotemporal Difference Booster (MSDB).

The MSDB integrates bitemporal image alignment to mitigate registration errors and a difference-concatenation dual-branch design to strengthen genuine change features, as illustrated in Fig. 5. Theoretically, this design exploits the complementarity between explicit difference modeling and implicit semantic reasoning. The Difference Branch captures macroscopic change salience to determine the scope, while the Concatenation Branch extracts structural priors to resolve the details. Consequently, this collaborative dual-branch design effectively suppresses non-substantive variations, exemplified

by seasonal fluctuations. Specifically, the MSDB employs deformable convolution (DeConv) to perform pixel-level alignment on H_{t2}^i , generating the spatially aligned feature $\hat{H}_{t2}^i$ with respect to H_{t1}^i . Building on this, the two parallel branches of the MSDB are designed with distinct mechanisms to achieve complementary functionality. The bitemporal image alignment process can be represented as

$$\begin{aligned}
\hat{H}_{t2}^i &= \text{DeConv}(H_{t2}^i) \\
H_{sub}^i &= |H_{t1}^i - \hat{H}_{t2}^i| \\
H_{con}^i &= \text{Concat}[H_{t1}^i, \hat{H}_{t2}^i]
\end{aligned} \quad (11)$$

where H_{sub}^i corresponds to the absolute difference of bitemporal features, and H_{con}^i indicates the channel-concatenated bitemporal features.

The difference branch takes H_{sub}^i as input and captures global change semantics (D_{sub}^i). The difference branch employs CA with a direction-aware global pooling strategy to effectively capture global difference information. It overcomes the limitations of fixed receptive fields inherent in conventional convolutions, thereby effectively capturing global changes within the image.

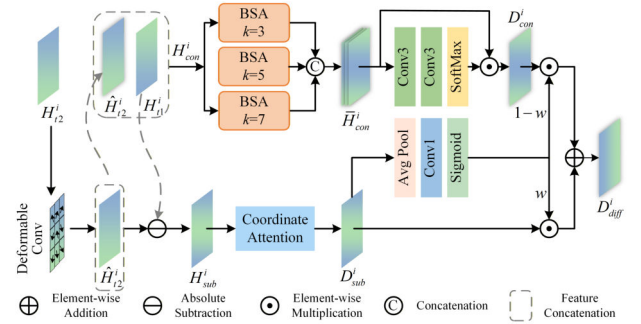


Fig. 5. Details of MSDB. The upper branch is the concatenation branch, and the lower branch is the difference branch.

The concatenation branch is designed to extract the model's sensitivity to fine-grained change details (D_{con}^i) and takes H_{con}^i as input. The core of this branch is the Bidirectional Strip Attention (BSA), which integrates the capability of convolutional operations to extract fine-grained change details with the adaptive weight allocation of attention mechanisms, as illustrated in Fig. 6. The BSA employs a dynamic kernel generator to produce attention kernels (Kh_k^i , Kw_k^i) for the horizontal and vertical orientations ($k \in \{3, 5, 7\}$). It adaptively captures salient features in strip regions, thereby improving the model's ability to capture fine-grained change details. For the horizontal computation, the input H_{con}^i is first transformed into an intermediate representation X_f^i through a feature refinement layer. Subsequently, X_f^i is processed by a dynamic kernel generator to produce an attention kernel $Kh_k^i \in \mathbb{R}^{h \times w \times c \times k}$. At the same time, X_f^i is reorganized into $Xh_k^i \in \mathbb{R}^{h \times w \times c \times k}$ via boundary padding and feature unfolding operations to match the shape of the attention kernel. Finally, the horizontal strip attention map $\text{Atth}_k^i \in \mathbb{R}^{h \times w \times c}$ is obtained by computing the inner product between Kh_k^i and Xh_k^i at each spatial location. The horizontal calculation process of BSA can be expressed as

$$\begin{aligned}
X_f^i &= \text{GELU}(\text{GN}(\text{Conv3}(H_{con}^i))) \\
Kh_k^i &= \delta(\text{Conv}_{1 \times k}^H(X_f^i)) \\
Xh_k^i &= \text{Unfold}_k^H(\text{Pad}_H(X_f^i)) \\
\text{Atth}_k^i &= \langle Kh_k^i, Xh_k^i \rangle
\end{aligned} \tag{12}$$

where GELU denotes the Gaussian Error Linear Unit activation function. $\text{Conv}_{1 \times k}^H$ represents the dynamic kernel generators for the horizontal direction. Pad_H denotes the boundary padding operations in the horizontal direction. Unfold_k^H corresponds to the feature unfolding operation in the horizontal direction, and $\langle \cdot, \cdot \rangle$ denotes the inner product operation. Similarly, the calculation process in the vertical direction is the same as the calculation process in the horizontal direction, generating a vertical strip attention map Attw_k^i . Atth_k^i and Attw_k^i are fused by element-wise summation to form the bidirectional strip attention feature $\text{BSA}_k^i \in \mathbb{R}^{h \times w \times c}$. Building on this mechanism, we construct three BSA modules with different receptive field sizes to capture local change details at multiple scales. The outputs of these modules are concatenated along the channel dimension to obtain $\bar{H}_{con}^i$. It is then processed by a convolution–Softmax fusion unit that adaptively highlights the most informative scale of change, yielding the output D_{con}^i .

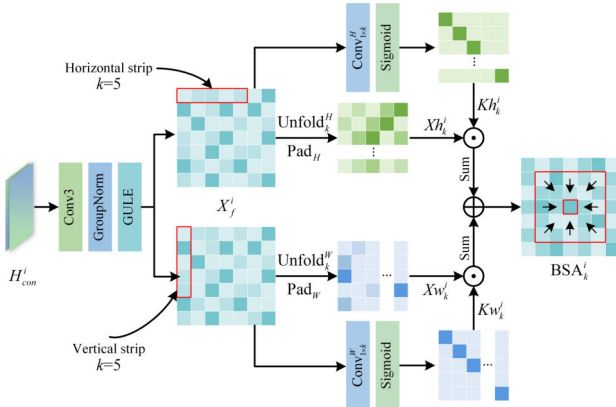


Fig. 6. Details of BSA.

D_{con}^i and D_{sub}^i from the two branches are fused through a weight generation mechanism composed of average pooling, a Conv1, and a Sigmoid function. The fusion process can be formulated as:

$$\begin{aligned}
G &= \delta(\text{Conv1}(\text{AvgPool}(D_{sub}^i))) \\
D_{diff}^i &= D_{sub}^i \odot G \oplus D_{con}^i \odot (1 - G)
\end{aligned} \tag{13}$$

In summary, MSDB effectively captures discriminative difference features of bitemporal RSI through the difference branch and the concatenation branch.

E. Adaptive Gated Attention Refiner (AGAR)

To bridge the semantic gap between deep-level semantics and shallow-level spatial details in the decoder, we propose the AGAR. As illustrated in Fig. 7, AGAR follows a three-stage paradigm, which is specifically designed to achieve a high-quality integration of cross-layer features.

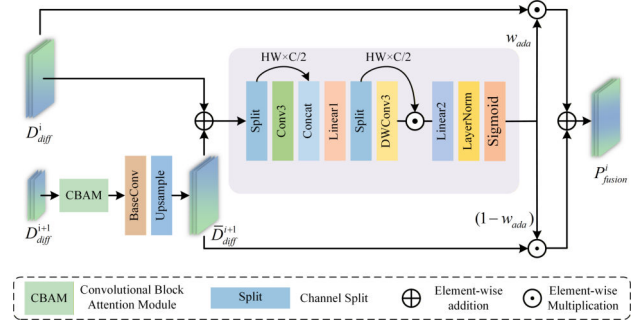


Fig. 7. Illustration of AGAR.

The AGAR module operates in three stages to refine and fuse multi-level features. First, deep features D_{diff}^{i+1} are enhanced via CBAM to suppress noise and then upsampled to align spatially with shallow features D_{diff}^i , yielding $\bar{D}_{diff}^{i+1}$. Next, the aligned deep and shallow features are processed via a gated fusion path involving channel-wise splitting and recombination. Consequently, adaptive weights w_{ada} are generated to bridge the semantic gap between feature levels. Finally, these weights are applied to fuse $\bar{D}_{diff}^{i+1}$ and D_{diff}^i in a content-aware manner to obtain the fused feature P_{fusion}^i , formulated as:

$$\begin{aligned}
P_{fusion}^i &= D_{diff}^i \odot w_{ada} \\
&\oplus \bar{D}_{diff}^{i+1} \odot (1 - w_{ada})
\end{aligned} \tag{14}$$

The proposed AGAR effectively bridges the semantic gap between high-level semantic features and low-level detail features and suppresses noise interference, which significantly improves change detection accuracy.

F. Loss Function

Since SSCD is inherently a pixel-wise binary classification task, we adopt BCE as the foundational loss function to optimize AMS-SemiCD. The total loss function $\mathcal{L}_{total}$ of the model comprises two components: a supervised loss $\mathcal{L}_{sup}$ computed on labeled data, and a semi-supervised consistency loss $\mathcal{L}_{unsup}$ calculated on unlabeled data.

Supervised Loss. We define the labeled data as (x, y) , where x denotes the input bitemporal image pair and y represents the pixel-wise change label. The supervised loss $\mathcal{L}_{sup}$ is defined as the Binary Cross-Entropy (BCE) between the prediction $P_s(x)$ of the student model and the ground truth label y :

$$\begin{aligned}
\mathcal{L}_{BCE}(\hat{Y}, Y) &= -(Y \log(\hat{Y}) + (1 - Y) \log(1 - \hat{Y})) \\
\mathcal{L}_{sup} &= \mathcal{L}_{BCE}(P_s(x), y)
\end{aligned} \tag{15}$$

where $\hat{Y}$ and Y denote the predicted change map and the ground truth label, respectively. Pixel values in y are binary $\{0, 1\}$, while those in $P_s(x)$ range continuously in $[0, 1]$, representing the probability of change versus no-change.

Semi-supervised Loss. For unlabeled data x' , we employ asymmetric augmentations to enforce consistency. Weak augmentation $\mathcal{A}_w$, comprising spatial transformations (e.g.,

flipping and rotation), is applied to the Teacher to yield stable targets $P_t(x'_w)$. For the Student, strong augmentation $\mathcal{A}_s$ builds upon $\mathcal{A}_w$ by further injecting photometric distortions (e.g., brightness and contrast) to produce predictions $P_s(x'_s)$. To enhance pseudo-label quality, we explicitly integrate the soft weight ω_i derived in (6) into the loss function. The weighting factor modulates the contribution of each pixel, ensuring the model focuses on reliable signals while suppressing the impact of noise:

$$\mathcal{L}_{unsup} = \frac{1}{N} \sum_{i=1}^N \omega_i \cdot \mathcal{L}_{BCE}(P_s(x'_s), P_t(x'_w)) \quad (16)$$

where $N = B \times H \times W$ denotes the total number of pixels in a training batch. By dynamically down-weighting unreliable samples via ω_i , the objective function effectively mitigates error accumulation from incorrect pseudo-labels.

Total Loss. Ultimately, the total loss function $\mathcal{L}_{total}$ of the model is the weighted sum of the two aforementioned losses:

$$\mathcal{L}_{total} = \mathcal{L}_{sup} + 0.2 \cdot \mathcal{L}_{unsup} \quad (17)$$

Following the Mean-Teacher model C2F [9], we set the weighting factor 0.2.

IV. EXPERIMENTS

In this section, we verify the effectiveness of our proposed method through extensive experiments, mainly including the introduction of the experimental conditions, ablation experiments, and comparison experiments.

A. Experimental Conditions

Datasets. GoogleGZ-CD [8] : A large-scale, very high-resolution (VHR) multispectral satellite image dataset openly released by Wuhan University. It encompasses 20 pairs of seasonal high-resolution images captured in the suburban areas of Guangzhou, China, spanning from 2006 to 2019. These images vary in dimensions from 1006×1168 pixels to 4936×5224 pixels, with a spatial resolution of 0.55 m. The dataset encompasses diverse image change types, such as water bodies, roads, farmland, bare land, forests, buildings, and vessels. For GPU training convenience, the original image pairs were cropped into non-overlapping blocks of 256×256 pixels. The resulting blocks were then randomly divided into 2,882 training, 360 validation, and 360 testing pairs, establishing an 8:1:1 custom split ratio.

LEVIR-CD [45] : A substantial VHR building change detection dataset made available by Beihang University. It comprises 637 pairs of high-resolution image blocks, captured from 2002 to 2018, with image dimensions of 1024×1024 pixels and a spatial resolution of 0.5 m. The dataset predominantly focuses on building-related changes and encompasses various types of structures, including villa residences, high-rise apartments, small garages, and large warehouses. To accommodate GPU memory limits and ensure equitable comparisons, the default image pairs were cropped into non-overlapping patches of 256×256 pixels. Following the official split setting,

this yielded 7,120 training, 1,024 validation, and 2,048 testing pairs.

WHU-CD [46] : WHU-CD comprises bitemporal high-resolution images with the size of $32,507 \times 15,345$. It is captured in the Christchurch region of New Zealand with a resolution of 0.2 m. It mainly focuses on building changes. For GPU training convenience, the original images were cropped into non-overlapping patches of 256×256 pixels. They were then randomly divided into 6,096 training, 762 validation, and 762 testing pairs, corresponding to an 8:1:1 custom split ratio.

In the semi-supervised experiments, we construct the datasets through strictly reproducible random sampling with a fixed seed of 42 to guarantee comparison fairness. We randomly select 5%, 10%, and 20% of the training data as labeled samples and treat the remaining data as unlabeled.

Implementation Details. To evaluate the proposed method, we implemented the model using the PyTorch framework. The Mamba branch and the CNN branch employ VMamba-small [47] and ResNeXt50 [48] as their backbones, respectively. All experiments were conducted on a single NVIDIA RTX 3090 GPU. The model was optimized using AdamW with a weight decay of 0.0025. We adopted a cosine annealing learning rate scheduler with an initial learning rate of $2e-4$. Due to GPU memory constraints, the batch size was set to 8. The model was trained for a total of 105 epochs to ensure convergence. For the first 5 epochs, training was conducted using only labeled samples. In the subsequent 100 epochs, all samples were utilized for training. To ensure a fair comparison, we utilized the official original datasets and performed data splitting with a fixed random seed. Consequently, all competing methods were retrained on these identical data splits. Meanwhile, we strictly followed the optimal data augmentation strategies, training epochs, and optimizer settings as specified in their respective original papers to guarantee their best performance.

Evaluation Metrics. To provide a comprehensive and objective assessment of the proposed model, we employ the F1 score (F1), overall accuracy (OA), Cohen's kappa coefficient (KC), and intersection over union (IoU) as evaluation metrics. The aforementioned indicators are computed in accordance with their standard formulations in the literature [9]. For all of these metrics, a higher value signifies superior change-detection performance.

Comparison Methods. In the experiments, we compared the following seven advanced SSCD methods in the past three years to validate the effectiveness of the proposed model:

1) SemiCD [49] revisits the semi-supervised consistency regularization paradigm. It enhances the model's generalization ability by enforcing prediction consistency constraints on both labeled and unlabeled data.

2) RCL [50] proposes a reliable contrastive learning mechanism. This approach enhances the model's robustness in distinguishing between changed and unchanged regions by designing a robust contrastive loss for positive and negative sample pairs.

3) FPA [51] introduces a dual mechanism that combines Feature Alignment (FA) and Prediction Alignment (PA). This approach significantly reduces intra-class variance while simultaneously reinforcing spatial consistency.

TABLE I
ABLATION STUDY OF KEY MODULES IN AMS-SemiCD ON THE LEVIR-CD DATASET

Method	5%				10%				20%			
	F1	IoU	OA	KC	F1	IoU	OA	KC	F1	IoU	OA	KC
Baseline	86.82	76.71	98.70	86.14	88.01	78.59	98.84	87.41	89.48	80.97	98.96	88.93
w/o ASPR	88.76	79.80	98.88	88.18	90.19	82.13	99.02	89.68	91.01	83.50	99.11	90.54
w/o MSSF	88.20	78.89	98.85	87.59	89.56	81.10	98.97	89.02	90.83	83.21	99.09	90.35
w/o MSDB	87.75	78.17	98.80	87.12	88.83	79.90	98.90	88.25	90.45	82.56	99.06	89.95
w/o AGAR	87.64	78.27	98.80	87.01	90.15	82.07	99.02	89.68	90.94	83.38	99.10	90.47
AMS-SemiCD	89.11	80.36 (+3.65)	98.92	88.66	90.61	82.84 (+4.25)	99.07	90.12	91.17	83.78 (+2.81)	99.12	90.17

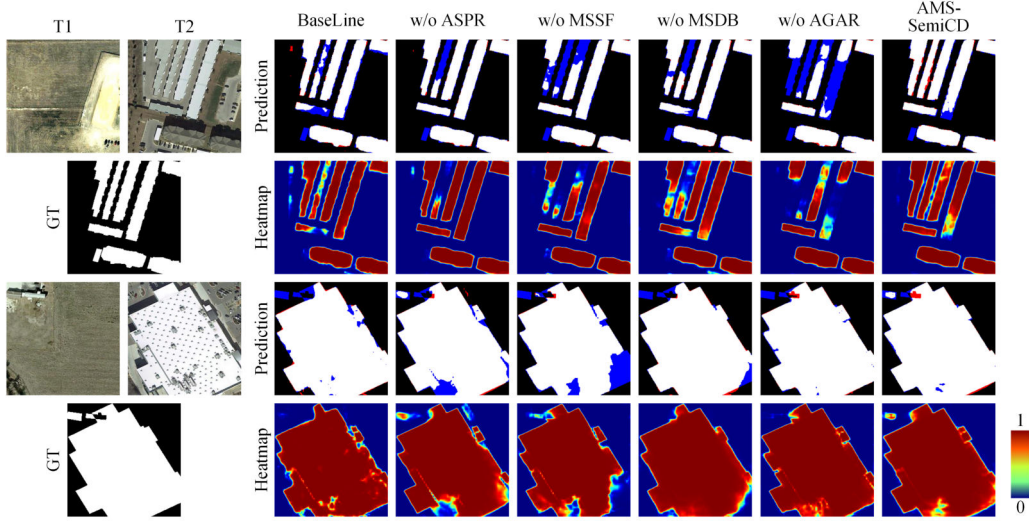


Fig. 8. Visualization results of the ablation study on the LEVIR-CD dataset with a 20% labeled ratio, showcasing the visual outputs of Baseline, w/o ASPR, w/o MSSF, w/o MSDB, w/o AGAR, and the complete AMS-SemiCD framework. T1 and T2 represent the bitemporal input images, and GT represents the ground truth. TP (white), TN (black), FP (red), and FN (blue).

4) C2F [9] is based on the Mean Teacher framework. It designs a Coarse-to-Fine multi-scale attention network (C2FNet) to progressively refine the extraction of change features.

5) UniMatch [33] transfers the weak-to-strong consistency regularization from semantic segmentation to CD. It operates by integrating prediction consistency under data augmentations of multiple strengths.

6) SemiCD-VL [40] innovatively employs a Visual-Language Model (VLM) to guide SSL. It generates high-quality pseudo-labels via a Change Event Generation strategy and enhances feature representations with contrastive regularization. Notably, SemiCD-VL is not an end-to-end model.

7) CutMix [35] integrates change-aware CutMix augmentation into the consistency regularization framework. It mitigates positive sample scarcity and change type imbalance by enforcing consistency on synthesized mixed samples.

The rationale for selecting these seven methods is as follows: SemiCD [49], FPA [51], C2F [9], UniMatch [33], and CutMix [35] are representative methods based on consistency regularization and are directly relevant to the research objec-

tives of this paper. RCL [50] employs contrastive learning to enhance feature representations, offering a distinct validation perspective. As a cutting-edge method, SemiCD-VL [40] utilizes foundation models to pre-generate high-quality pseudo-labels. Comparison against these mainstream and advanced benchmarks serves to fully validate the effectiveness and superiority of our proposed method.

B. Ablation Experiments

To validate the effectiveness of the key components within our proposed AMS-SemiCD framework, we conducted a series of ablation experiments. Specifically, we evaluated the contributions of ASPR, MSSF, MSDB, and AGAR. The experiments were performed under varying semi-supervised settings, using 5%, 10%, and 20% of the labeled data.

1) Effectiveness of proposed modules

a) Component-Wise Ablation Analysis

The numerical results of our leave-one-out ablation study are detailed in Table I. We started with the Baseline method and systematically removed each key component to observe

TABLE II
STEP-WISE ABLATION STUDY EVALUATING THE INCREMENTAL CONTRIBUTION OF ASPR, MSSF, MSDB, AND AGAR ON THE LEVIR-CD DATASET

ASPR	MSSF	MSDB	AGAR	5%			10%			20%		
				F1	IoU	KC	F1	IoU	KC	F1	IoU	KC
✗	✗	✗	✗	86.82	76.71	86.14	88.01	78.59	87.41	89.48	80.97	88.93
✓	✗	✗	✗	87.30	77.47	86.64	89.14	80.41	88.58	90.30	82.32	89.80
✓	✓	✗	✗	87.63	77.99	86.99	90.07	81.93	89.54	90.77	83.10	90.28
✓	✓	✓	✗	87.64	78.27	87.01	90.15	82.07	89.68	90.94	83.38	90.47
✓	✓	✓	✓	89.11	80.36	88.66	90.61	82.84	90.12	91.17	83.78	90.17

the impact on performance. As the results indicate, the full AMS-SemiCD method consistently achieves the best performance across all label ratios, demonstrating the synergistic effect of all proposed modules. For instance, under the 10% label setting, the full method achieves an F1 score of 90.61% and an IoU of 82.84%. Removing the ASPR (w/o ASPR) leads to the most significant performance degradation, with the F1 score dropping to 90.19% and IoU to 82.13%. This highlights the critical role of the ASPR strategy in refining pseudo-labels and increasing the use of unlabeled data. Similarly, removing the MSSF (w/o MSSF), MSDB (w/o MSDB), or AGAR (w/o AGAR) modules also results in noticeable performance drops compared to the complete AMS-SemiCD method. The ablation results verify that each component provides a valuable contribution to the final change detection results.

The qualitative evaluation on the LEVIR-CD dataset under a 20% labeled ratio in Fig. 8 visually corroborates the distinct roles of each module, consistent with the quantitative metrics presented in Table I. Specifically, removing ASPR (w/o ASPR) leads to significant omissions of subtle changes, such as the small buildings in row 1. The deficiency confirms its necessity for effectively mining latent information from medium-to-low confidence samples. Notably, 'w/o MSSF' exhibits severe internal fragmentation within large objects, as illustrated by the roof holes in row 2. The fragmentation appears even more structurally incoherent than the Baseline. This visual degradation underscores that without MSSF's synergistic global-local fusion, the model fails to maintain object integrity. The 'w/o MSDB' column displays blurred and irregular boundaries, validating the module's role in refining edge precision via the synergy of explicit difference cues and structural priors. Finally, 'w/o AGAR' suffers from artifacts, highlighting the need for effective cross-layer fusion. In contrast, the full AMS-SemiCD generates coherent, noise-free maps with sharp boundaries, intuitively demonstrating the synergistic efficacy of the proposed components

b) Step-Wise Integration Analysis

We conducted a step-wise ablation study to scrutinize the synergistic contribution of each component within AMS-SemiCD. Specifically, we incrementally integrated ASPR, MSSF, MSDB, and AGAR into the base network. The quantitative results on the LEVIR-CD dataset are summarized in Table II. Starting with the pure baseline (without any proposed modules), the model yields an F1 score of 86.82%

and IoU of 76.71% at a 5% label ratio. Incorporating the ASPR strategy immediately boosts the IoU to 77.47%, validating the importance of refining pseudo-labels for effective semi-supervised learning. The subsequent introduction of MSSF further improves the IoU to 77.99%, confirming the benefit of fusing global and local features. Subsequently, adding MSDB boosts the IoU from 83.10% to 83.38% at the 20% ratio, validating its efficacy in improving change detection accuracy. Finally, the integration of AGAR triggers a substantial performance leap, ultimately boosting the IoU score to 80.36% at the 5% ratio. The significant increase highlights the necessity of effective cross-layer fusion to reduce artifacts.

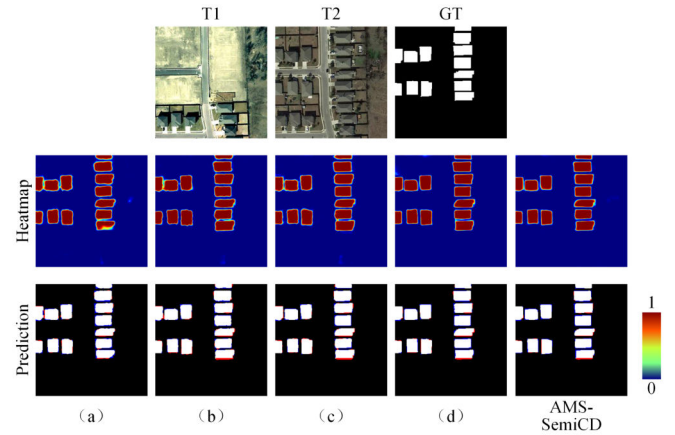


Fig. 9. Progressive visualization results of the step-wise ablation study on the LEVIR-CD dataset with a 20% labeled ratio. (a) Baseline. (b) Baseline + ASPR. (c) Baseline + ASPR + MSSF. (d) Baseline + ASPR + MSSF + MSDB. AMS-SemiCD represents the complete framework incorporating the AGAR module. T1 and T2 represent the bitemporal input images, while GT indicates the ground truth. In the prediction maps, white, black, red, and blue regions correspond to TP, TN, FP, and FN, respectively.

We illustrate the corresponding visualization results on the LEVIR-CD dataset with a 20% labeled ratio in Fig. 9, which intuitively corroborate the quantitative findings in Table II. The baseline model generates low-confidence heatmaps and exhibits severe boundary misclassifications indicated by prominent red false-positive regions. Integrating the ASPR strategy successfully recovers missing structural details through the effective mining of soft-label information. Sequentially adding the MSSF and MSDB modules systematically suppresses the prominent false positives to refine local object boundaries.

Ultimately, the complete AMS-SemiCD framework equipped with the AGAR module generates highly confident heatmaps and achieves optimal visual consistency with the ground truth.

Finally, to validate the generalizability of our proposed components across different scenarios, we supplemented the step-wise ablation studies on the GoogleGZ-CD and WHU-CD datasets with a 5% labeled ratio. We present these results using radar charts in Fig. 10 to provide a more intuitive multi-metric comparison. As depicted by the overall expansion of the polygons, the progressive integration of our modules significantly improves the target-sensitive metrics (F1, IoU, and KC) on both datasets under this sparse supervision setting. Notably, on the GZ dataset, we observe a slight fluctuation in OA during intermediate stages (e.g., +ASPR and +MSDB). Since change detection datasets suffer from extreme class imbalance, OA is heavily dominated by the vast unchanged background. This temporary minor drop indicates that intermediate modules aggressively mine foreground pseudo-labels to recover missing changes, which inevitably introduces slight background noise. Crucially, the final integration of the AGAR module effectively refines these boundaries and suppresses false positives, restoring and maximizing the OA. This consistent trajectory of improvement aligns perfectly with our observations on LEVIR-CD, robustly proving that the complete framework is highly generalizable and its internal components are intrinsically complementary.

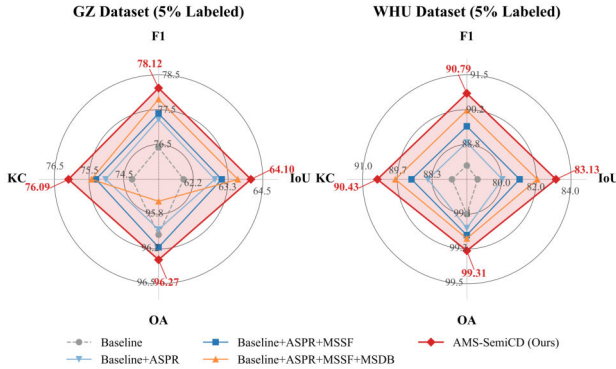


Fig. 10. Radar charts illustrating the step-wise ablation study results on the (a) GoogleGZ-CD and (b) WHU-CD datasets with a 5% labeled ratio. The outward expansion of the polygons intuitively demonstrates the comprehensive performance gains across all four metrics (F1, IoU, OA, and KC) achieved by progressively integrating ASPR, MSSF, MSDB, and AGAR.

2) Effectiveness of the Dual-Branch Synergistic Design

To verify the necessity of integrating CNN and Mamba, we compare single-branch variants against our dual-branch architecture on the WHU-CD dataset using 5% labeled samples. The quantitative performance and computational complexity metrics are summarized in Table III. As observed, the CNN-only baseline yields the lowest F1 score of 77.69%, restricted by its limited receptive field. In contrast, the Mamba-only variant significantly improves performance to 87.53% F1 while maintaining a low computational cost of 30.86 GFLOPs, demonstrating the efficiency of global context modeling. However, relying solely on global modeling may overlook fine-grained local details. Crucially, the proposed dual-branch design achieves the optimal performance, reaching an F1 score

of 90.79% and an IoU of 83.13%. This represents a substantial improvement of 3.26% in F1 over the Mamba-only version and 13.10% over the CNN-only version. Although the dual-branch integration increases model size and computation, complementary local-global modeling substantially improves F1 and IoU, indicating that the additional complexity is accompanied by clear accuracy gains.

TABLE III
COMPARISON OF SINGLE-BRANCH AND DUAL-BRANCH ARCHITECTURES ON THE WHU-CD DATASET (5% LABELED RATIO)

CNN	Mamba	F1	IoU	FLOPs/G	Params/M
✓	✗	77.69	63.52	41.16	159.95
✗	✓	87.53	77.83	30.86	184.61
✓	✓	90.79	83.13	53.69	240.90

3) Effectiveness of ASPR Strategy

We investigate the contributions of the Class-adaptive Dynamic Threshold and Contribution-adaptive Soft Weighting via a detailed ablation study on the WHU-CD dataset. Additionally, we verify the superiority of the dynamic mechanism over fixed hyperparameters. The results are reported in Table IV.

TABLE IV
ABLATION STUDY OF CLASS-ADAPTIVE DYNAMIC THRESHOLD AND CONTRIBUTION-ADAPTIVE SOFT WEIGHTING ON WHU-CD DATASET

Dyn. Thr.	Soft Wgt.	5%		20%	
		F1	IoU	F1	IoU
✗	✗	88.03	78.62	91.55	84.41
✓	✗	89.75	81.40	92.50	86.05
0.60	✓	88.58	79.50	92.11	85.38
0.80	✓	90.47	82.60	92.57	86.17
0.95	✓	90.14	82.05	91.91	85.03
✓	✓	90.79	83.13	93.38	87.58

Regarding individual effects, incorporating the Dynamic Threshold alone yields consistent gains across distinct label ratios. Specifically, at the 5% ratio, it increases the F1 score from 88.03% to 89.75% and IoU from 78.62% to 81.40%, confirming its capability to robustly filter noise even under scarce supervision. To justify its design against hyperparameter selection concerns, we compared the dynamic strategy with fixed thresholds ($\tau_t^{(c)} = 0.60, 0.80, 0.95$) combined with soft weighting. As observed, fixed thresholds exhibit notable performance fluctuations, indicating high sensitivity to manual tuning. In contrast, our Dynamic Threshold strategy eliminates this dependency, surpassing even the optimally tuned fixed threshold ($\tau_t^{(c)} = 0.80$). Ultimately, the full ASPR strategy achieves peak performance, reaching an F1 of 93.38% and IoU of 87.58% at the 20% ratio. The results demonstrate the effectiveness of the proposed ASPR strategy. Here, the dynamic threshold serves as a reliable anchor for soft weighting to maximally leverage latent information.

4) Effectiveness of the EFI Module

To address the concern regarding the complexity and design motives of the MSSF structure, we evaluated the necessity of the EFI module. We conducted a comparative experiment by replacing the EFI module with a standard convolutional block consisting of two stacked 3×3 convolutions, BN, and ReLU layers. The primary objective of this analysis is to verify whether the performance gains stem from the specific attention-gating design or merely from increased model capacity. The quantitative results are presented in Table V.

TABLE V
PERFORMANCE AND COMPLEXITY COMPARISON BETWEEN THE PROPOSED EFI MODULE AND A SIMPLIFIED CONVOLUTIONAL BASELINE ON THE WHU-CD DATASET (5% LABELED RATIO)

EFI	F1	IoU	FLOPs/G	Params/M
✗	87.23	77.36	72.15	288.25
✓	90.79	83.13	53.69	240.90

As observed, the simplified variant yields an F1 score of 87.23% and an IoU of 77.36%. Surprisingly, this simplified design incurs higher computational costs, requiring 72.15 GFLOPs and 288.25M parameters. In contrast, the proposed EFI module achieves a superior F1 score of 90.79% and an IoU of 83.13%. Crucially, the inclusion of EFI reduces the computational burden to 53.69 GFLOPs and the parameter count to 240.90M compared to the stacked convolution baseline. The result conclusively demonstrates that the superimposed components within EFI, such as coordinate attention and gating, function as efficient mechanisms for feature refinement. Rather than artificially inflating model capacity, the EFI design is indispensable for achieving high accuracy while effectively reducing computational redundancy.

5) Visualization of Pseudo-label Evolution

To intuitively verify the effectiveness of the ASPR strategy, we analyze pseudo-label evolution on GoogleGZ-CD with a 20% labeled ratio at Epochs 10, 50, and 100, as shown in Fig. 11. At Epoch 10, raw predictions contain substantial noise, which ASPR suppresses to reduce error accumulation. This conservative refinement may also attenuate low-confidence change signals and cause missed detections. By Epoch 50, refined pseudo-labels exhibit clearer boundaries and improved structural integrity as object representations become more reliable. At Epoch 100, most predictions are stable, although some extremely low-confidence fine-grained details remain suppressed. This behavior reflects ASPR's preference for reliable supervision, which improves training stability but may sacrifice certain hard examples.

6) Parameter Sensitivity Analysis

We analyzed the sensitivity of the initial scaling factor ξ_{init} and momentum m on the WHU-CD dataset at a 5% labeled ratio. The validation F1 score was recorded to determine whether early-stage instability causes suboptimal EMA convergence. A small momentum ($m = 0.99$) updates these statistical weights too rapidly, making the framework susceptible to early noisy pseudo-labels and causing severe validation fluctuations, as illustrated in Fig. 12(a). Furthermore, Fig. 12(b) reveals

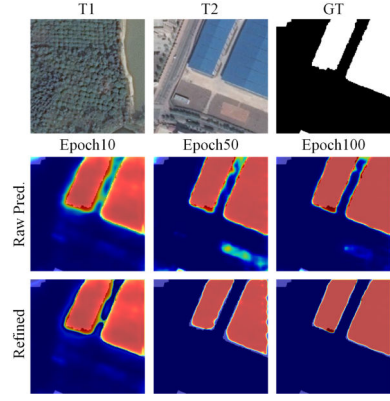


Fig. 11. Visualization of pseudo-label evolution during training on the GoogleGZ-CD dataset with a 20% labeled ratio. The middle and bottom rows compare the Raw Predictions (Teacher output) with the Refined Pseudo-labels (ASPR output) at Epochs 10, 50, and 100. The visualization demonstrates how ASPR suppresses noise in the early stages and refines object boundaries as training converges.

the risks of an overly large scaling factor ($\xi_{init} = 3$). It over-amplifies unstable early signals, forcing the network to memorize erroneous predictions and converge to a suboptimal state. In contrast, our default setting ($m = 0.999$, $\xi_{init} = 2$) guarantees a smooth and stable convergence trajectory, effectively mitigating the risk of early-stage instability.

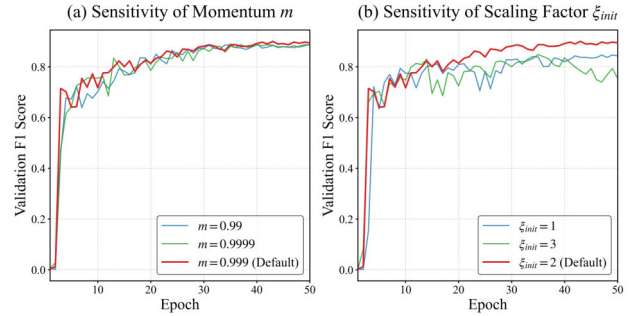


Fig. 12. Convergence curves of the validation F1 score on the WHU-CD dataset with a 5% labeled ratio. Sensitivity analysis of (a) momentum m and (b) initial scaling factor ξ_{init} . Red curves represent our default settings ($m = 0.999$, $\xi_{init} = 2$), which ensure stable convergence and optimal performance while effectively mitigating early-stage statistical instability.

The quantitative results in Table VI further corroborate these dynamic observations under the 5% labeled setting. Specifically, our default configuration achieves the highest performance across all metrics, peaking at an F1 score of 90.79% and an IoU of 83.13%. Any deviation from this optimal balance leads to noticeable performance degradation. For instance, setting $\xi_{init} = 3$ sharply drops the F1 score to 87.39%, explicitly confirming the detrimental impact of early-stage instability. Similarly, altering the ASPR momentum m to 0.99 or 0.9999 yields suboptimal F1 scores of 89.97% and 90.24%, respectively. Ultimately, the empirical data rigorously demonstrate that our selected hyperparameters provide the optimal balance for robust semi-supervised change detection.

7) Comparison with Pseudo-Labeling Strategies

To further distinguish ASPR from existing adaptive pseudo-labeling strategies, we conduct a controlled comparison un-

TABLE VI
SENSITIVITY ANALYSIS OF THE MOMENTUM PARAMETER m AND THE INITIAL SCALING FACTOR ξ_{init} ON THE WHU-CD DATASET (5% LABEL RATIO)

m	ξ_{init}	F1	IoU	KC
0.99	2	89.97	81.77	89.57
0.9999	2	90.24	82.22	89.86
0.999	3	87.39	77.60	86.89
0.999	1	86.75	76.60	86.23
0.999	2	90.79	83.13	90.43

der the 5% labeled setting on WHU-CD. All variants use the same Teacher–Student framework, MSE-Net architecture, data split, augmentation, and optimization settings, with only the pseudo-labeling strategy replaced. FlexMatch-based, FreeMatch-based, and SoftMatch-based denote implementations of their respective thresholding or weighting mechanisms within the same framework.

Adaptive thresholding improves pseudo-label learning over the fixed-threshold baseline, as shown in Table VII. FlexMatch-based and FreeMatch-based increase IoU by 1.87% and 0.93%, respectively, confirming the effectiveness of threshold adaptation. The SoftMatch-based strategy achieves an IoU of 77.92%, which is 0.70% lower than the fixed-threshold baseline. This result suggests that global confidence statistics may be insufficient to characterize the distinct confidence distributions of changed and unchanged pixels in this SSCD setting. ASPR achieves the highest F1 and IoU of 90.79% and 83.13%, outperforming the best alternative, FlexMatch-based, by 1.60% and 2.64%, respectively. The results demonstrate the effectiveness of jointly deriving class-adaptive dynamic thresholds and contribution-adaptive soft weights from class-wise confidence statistics.

TABLE VII
COMPARISON OF DIFFERENT PSEUDO-LABELING STRATEGIES ON THE WHU-CD DATASET UNDER THE 5% LABELED SETTING

Strategy	F1 (%)	IoU (%)
Fixed Threshold	88.03	78.62
FlexMatch-based	89.19	80.49
FreeMatch-based	88.61	79.55
SoftMatch-based	87.59	77.92
ASPR	90.79	83.13

8) Intrinsic Synergy Between MSE-Net and ASPR

To examine the intrinsic synergy between MSE-Net and ASPR, we compare MSE-Net with the representative BIT architecture on WHU-CD under the 5% labeled setting, as illustrated in Fig. 13. Both architectures are evaluated with three progressive pseudo-labeling strategies: a fixed hard threshold of 0.95, a class-adaptive dynamic threshold with hard selection, and the full ASPR. Replacing the fixed threshold with ASPR improves the IoU of BIT by 1.98% (71.51% to 73.49%),

confirming its effectiveness with an alternative architecture. Applying ASPR to MSE-Net yields a larger gain of 3.87% (79.26% to 83.13%). The larger improvement indicates that the robust feature representations of MSE-Net better support the class-wise statistical estimation and adaptive weighting of ASPR.

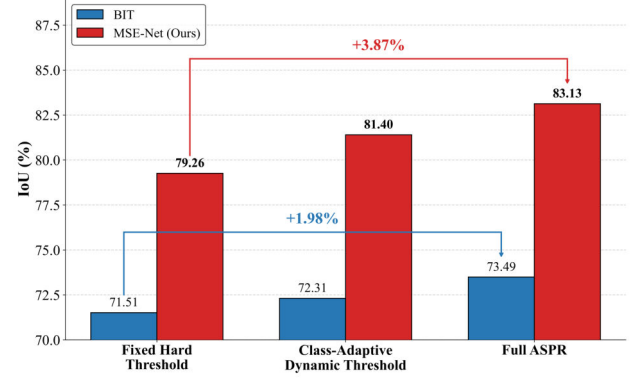


Fig. 13. IoU performance of BIT and MSE-Net on the WHU-CD dataset with a 5% labeled ratio under three progressive pseudo-labeling configurations: Fixed Hard Threshold of 0.95, Class-Adaptive Dynamic Threshold, and Full ASPR. The significantly larger IoU gain achieved by MSE-Net (+3.87%) compared to BIT (+1.98%) explicitly demonstrates the strong intrinsic synergy between our proposed architecture and the complete semi-supervised strategy.

C. Comparison Experiments

1) Comparative Experiments on GoogleGZ-CD Dataset

The quantitative results of our proposed AMS-SemiCD method against other advanced SSCD methods on the GoogleGZ-CD dataset are presented in Table VIII. The results show that AMS-SemiCD achieves the best performance at the 5% and 20% labeled ratios, and obtains tied-best F1 and IoU at the 10% labeled ratio. This demonstrates the strong competitiveness and robustness of the proposed method under different semi-supervised settings.

With only 5% of labeled samples, AMS-SemiCD achieved an F1 score of 78.12%, an IoU of 64.10%, and a KC of 76.09%, exceeding the second-best method by 2.57%, 3.39%, and 2.6%, respectively. As the proportion of labeled data increased to 20%, AMS-SemiCD reaches an IoU of 73.91%, significantly outperforming the latest CutMix (62.59%) and SemiCD-VL (68.62%). It is noteworthy that most change detection datasets inherently exhibit class imbalance, often resulting in suboptimal KC performance. The superior KC scores further validate the model's robustness in handling class imbalance inherent in change detection tasks.

For the qualitative evaluation, a comparative visual analysis is conducted using a 20% labeling ratio, as illustrated in Fig. 14. The examples in Fig. 14 (a-c) highlight the superiority of AMS-SemiCD in maintaining object integrity. While competing methods like CutMix often suffer from incomplete target structures, our method preserves complete change regions with precise boundaries. Furthermore, the results in Fig. 14 (d-f) demonstrate robust noise suppression capabilities. As observed, other approaches struggle with extensive red artifacts caused by complex backgrounds, whereas AMS-SemiCD

TABLE VIII
QUANTITATIVE PERFORMANCE OF DIFFERENT METHODS ON THE GOOGLEGZ-CD DATASET
(THE BEST VALUES ARE BOLD, AND THE SECOND BEST VALUES ARE UNDERLINED)

Method	5%				10%				20%			
	F1	IoU	OA	KC	F1	IoU	OA	KC	F1	IoU	OA	KC
SemiCD [49]	63.26	46.26	94.05	60.08	72.84	57.28	95.03	70.10	78.11	64.08	96.35	76.14
RCL [50]	72.41	56.75	95.39	69.91	75.61	62.09	95.87	74.35	80.26	67.03	96.53	78.36
FPA [51]	75.44	60.56	<u>96.24</u>	<u>73.49</u>	74.84	59.79	95.97	72.69	82.50	70.22	97.02	80.89
UniMatch [33]	66.37	49.67	95.16	63.97	78.98	65.26	96.45	77.05	82.65	70.43	97.11	81.09
C2F [9]	73.66	58.30	95.86	71.48	79.63	66.15	96.60	77.79	<u>84.17</u>	<u>72.67</u>	<u>97.30</u>	<u>82.70</u>
SemiCD-VL [40]	<u>75.55</u>	<u>60.71</u>	95.77	73.24	<u>79.80</u>	<u>66.39</u>	96.72	78.05	81.39	68.62	96.95	79.75
CutMix [35]	73.53	58.14	95.49	71.08	76.57	62.04	96.14	74.49	76.99	62.59	96.22	74.69
AMS-SemiCD	78.12	64.10	96.27	76.09	79.80	66.39	<u>96.63</u>	<u>77.97</u>	85.00	73.91	97.44	83.61
Fully-sup	F1 = 90.04, IoU = 81.88, OA = 98.23, KC = 89.07											

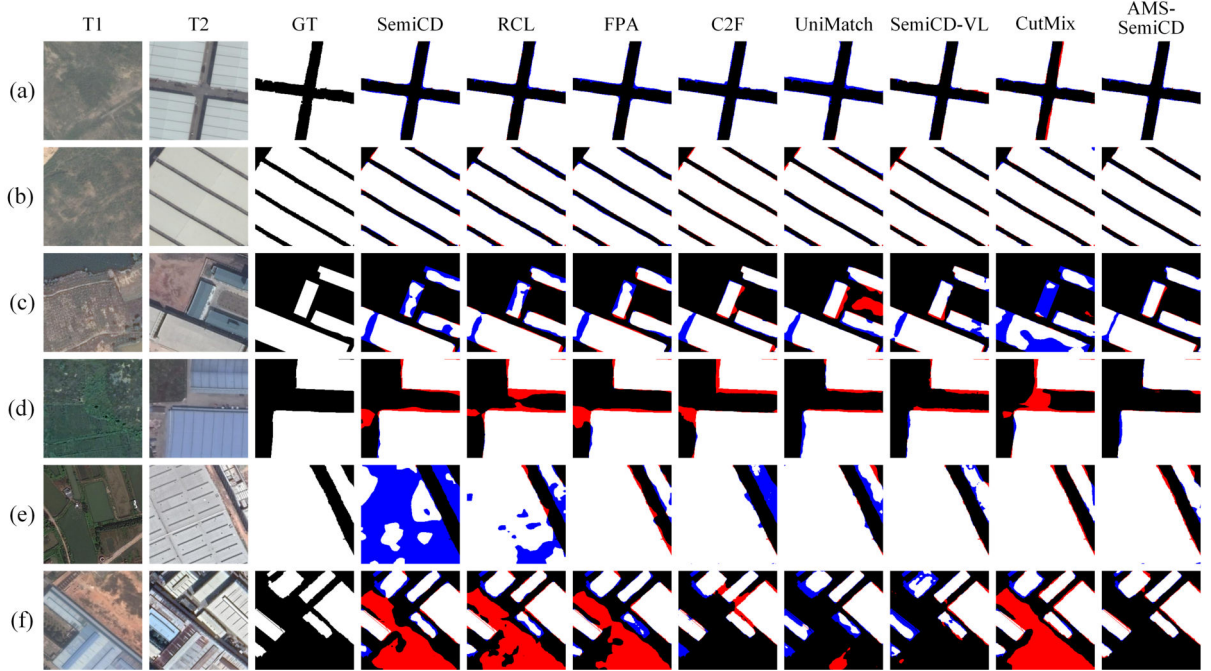


Fig. 14. Qualitative evaluation results of other advanced SSCD methods with a 20% labeled ratio on the GoogleGZ-CD dataset. TP (white), TN (black), FP (red), and FN (blue).

effectively filters out such interference, yielding cleaner and more reliable change maps.

2) Comparative Experiments on WHU-CD Dataset

The quantitative performance of our proposed AMS-SemiCD method against other advanced SSCD methods on the WHU-CD dataset is detailed in Table IX. The results demonstrate that AMS-SemiCD consistently achieves superior performance across all label ratios (5%, 10%, and 20%) in terms of F1, IoU, and KC.

With only 5% of labeled samples, AMS-SemiCD achieved an F1 score of 90.79%, an IoU of 83.13%, and a KC of 90.43%, surpassing the second-best method by margins of

1.41%, 2.33%, and 1.47%, respectively. As the proportion of labeled data increased to 20%, AMS-SemiCD reaches an IoU of 87.58%, significantly outperforming the latest CutMix (80.98%) and SemiCD-VL (83.86%). The consistently higher KC scores further attest to the model's stability and robustness in distinguishing changing foregrounds from complex backgrounds compared to existing advanced approaches.

For the qualitative evaluation, a comparative visual analysis is conducted using a 20% labeling ratio, as illustrated in Fig. 15. The results in Fig. 15 (a-d, f) reveal that competing methods are highly susceptible to background interference. Consequently, they generate extensive false alarms across di-

TABLE IX
QUANTITATIVE PERFORMANCE OF DIFFERENT METHODS ON THE WHU-CD DATASET
(THE BEST VALUES ARE BOLD, AND THE SECOND BEST VALUES ARE UNDERLINED)

Method	5%				10%				20%			
	F1	IoU	OA	KC	F1	IoU	OA	KC	F1	IoU	OA	KC
SemiCD [49]	77.57	63.36	98.30	76.68	80.06	66.75	98.41	79.24	82.90	70.79	98.67	82.21
RCL [50]	80.45	67.30	98.65	79.76	82.08	69.60	98.77	81.45	87.36	77.56	99.05	86.87
FPA [51]	83.52	71.70	98.74	82.86	87.85	78.34	99.10	87.39	87.96	78.50	99.08	87.48
UniMatch [33]	<u>89.38</u>	<u>80.80</u>	<u>99.18</u>	<u>88.96</u>	89.67	81.27	99.22	89.26	90.13	82.03	99.24	89.73
C2F [9]	85.97	75.40	98.93	85.42	88.42	79.25	99.09	87.95	90.73	83.03	99.29	90.36
SemiCD-VL [40]	88.42	79.24	99.17	87.99	<u>90.29</u>	<u>82.29</u>	<u>99.29</u>	<u>89.92</u>	<u>91.22</u>	<u>83.86</u>	<u>99.34</u>	<u>90.88</u>
CutMix [35]	87.27	77.42	99.04	86.77	87.49	77.77	99.06	87.01	89.49	80.98	99.21	89.08
AMS-SemiCD	90.79	83.13	99.31	90.43	91.64	84.57	99.38	91.32	93.38	87.58	99.51	93.12
Fully-sup:	F1 = 94.59, IoU = 89.73, OA = 99.60, KC = 94.38											

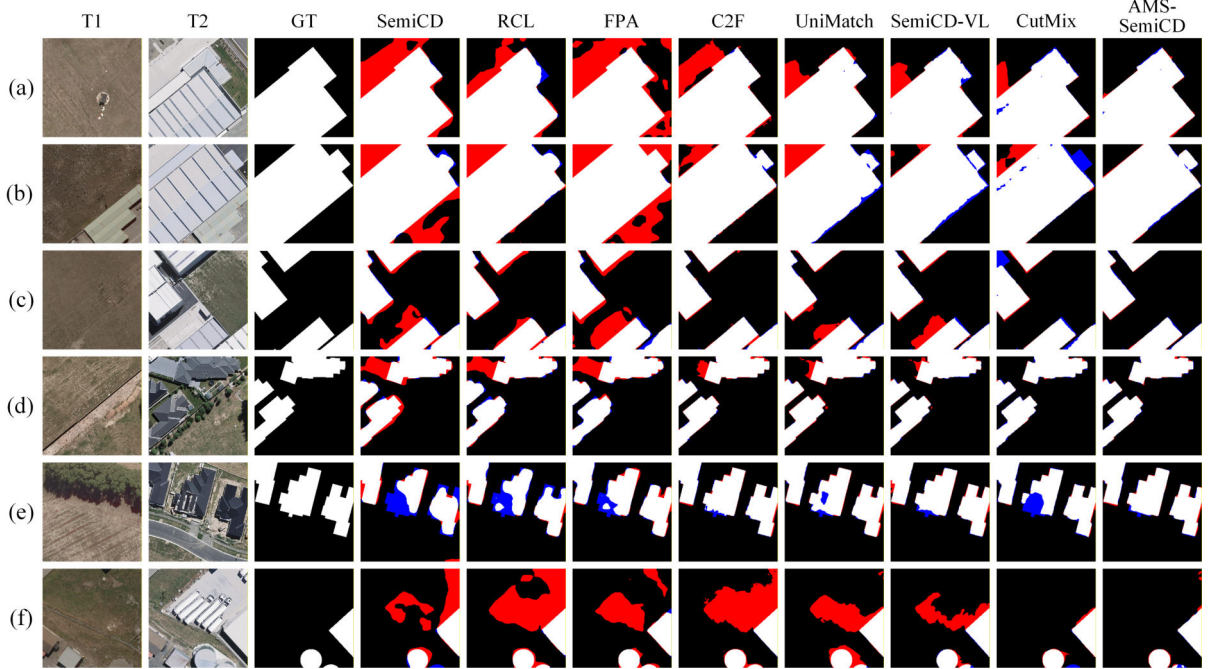


Fig. 15. Qualitative evaluation results of other advanced SSCD methods with a 20% labeled ratio on the WHU-CD dataset. TP (white), TN (black), FP (red), and FN (blue).

verse targets, ranging from large warehouses to irregular small buildings. Conversely, the structural complexity of buildings hinders competing methods, leading to fragmented predictions and significant internal omissions, as shown in Fig. 15 (e). In contrast, AMS-SemiCD effectively mitigates background interference while preserving the completeness of complex instances, thereby yielding significantly cleaner and more reliable change maps.

3) Comparative Experiments on LEVIR-CD Dataset

To further validate the performance of our AMS-SemiCD method, we conducted extensive comparative experiments on the LEVIR-CD dataset using identical baselines and evaluation

metrics. The results are presented in Table X.

With only 5% of labeled samples, AMS-SemiCD achieves an F1 score of 89.11%, an IoU of 80.36%, and a KC of 88.66%, surpassing all comparison methods comprehensively. At 20% labeled data, AMS-SemiCD achieves an IoU of 83.78%, surpassing the latest CutMix (75.17%) by a significant margin of 8.61%. Notably, our method approaches the fully supervised upper bound with a negligible F1 deficit of only 0.87%, demonstrating exceptional data efficiency.

For the qualitative evaluation, a comparative visual analysis is conducted using a 20% labeling ratio, as illustrated in Fig. 16. The results in Fig. 16 (a-b) reveal that comparison

TABLE X
QUANTITATIVE PERFORMANCE OF DIFFERENT METHODS ON THE LEVIR-CD DATASET
(THE BEST VALUES ARE BOLD, AND THE SECOND BEST VALUES ARE UNDERLINED)

Method	5%				10%				20%			
	F1	IoU	OA	KC	F1	IoU	OA	KC	F1	IoU	OA	KC
SemiCD [49]	85.18	74.19	98.58	84.45	87.08	77.12	98.74	86.42	87.55	77.85	98.79	86.91
RCL [50]	84.45	73.09	98.46	83.65	85.34	74.43	98.55	84.58	86.03	75.48	98.60	85.29
FPA [51]	84.60	73.32	98.53	83.84	86.89	76.81	98.72	86.22	87.05	77.06	98.75	86.39
UniMatch [33]	87.89	78.40	98.85	87.29	89.59	81.15	98.98	89.06	89.62	81.19	98.75	89.09
C2F [9]	<u>88.87</u>	<u>79.97</u>	<u>98.87</u>	<u>88.27</u>	90.24	82.22	99.02	89.72	<u>90.52</u>	<u>82.68</u>	<u>99.05</u>	<u>90.02</u>
SemiCD-VL [40]	88.53	79.42	98.74	87.94	<u>90.51</u>	<u>82.66</u>	<u>99.05</u>	<u>90.01</u>	90.10	81.99	99.00	89.58
CutMix [35]	84.54	73.22	98.51	83.76	85.58	74.79	98.60	84.85	85.83	75.17	98.62	85.10
AMS-SemiCD	89.11	80.36	98.92	88.66	90.61	82.84	99.07	90.12	91.17	83.78	99.12	90.17
Fully-sup: F1 = 92.04, IoU = 85.25, OA = 99.20, KC = 91.62												

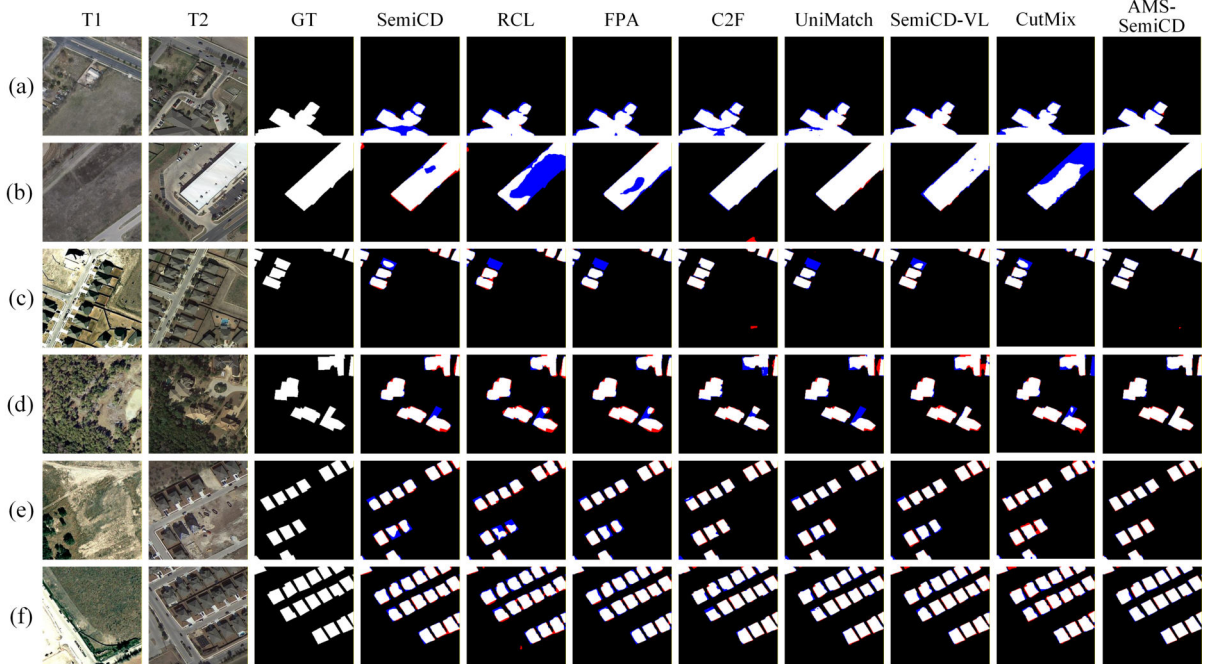


Fig. 16. Qualitative evaluation results of other advanced SSCD methods with a 20% labeled ratio on the LEVIR-CD dataset. TP (white), TN (black), FP (red), and FN (blue).

methods struggle with object integrity. Consequently, they generate fragmented predictions with significant internal omissions, particularly evident in large structures. On the other hand, the dense arrangement of small buildings hinders competing methods, leading to extensive edge artifacts and false alarms due to background interference, as shown in Fig. 16 (c-f). In contrast, AMS-SemiCD effectively mitigates these issues while preserving the completeness of instances and delineating precise contours, consequently producing highly accurate prediction maps.

D. Stability Analysis with Multiple Random Seeds

To evaluate the stability and statistical reliability of AMS-SemiCD, we conducted additional experiments with three independent random runs under the 5% labeled setting. Run 1 uses the default random seed 42 adopted in the main experiments, while Runs 2 and 3 use additional random seeds 43 and 44, respectively. AMS-SemiCD achieves mean IoU values of 64.80%, 80.13%, and 82.81% on GoogleGZ-CD, LEVIR-CD, and WHU-CD, respectively, as shown in Table XI. The corresponding standard deviations are 0.62, 0.42, and 0.31, indicating limited performance fluctuations

across different random seeds. These results demonstrate that AMS-SemiCD maintains stable performance under different random initializations and data samplings.

TABLE XI
STABILITY ANALYSIS OF AMS-SEMICD UNDER DIFFERENT RANDOM SEEDS WITH A 5% LABELED RATIO. THE IOU VALUES ARE REPORTED IN PERCENTAGE.

Dataset	Run 1	Run 2	Run 3	Mean	Std.
GoogleGZ-CD	64.10	65.05	65.26	64.80	0.62
LEVIR-CD	80.36	80.38	79.65	80.13	0.42
WHU-CD	83.13	82.80	82.51	82.81	0.31

Note: Run 1 uses the default seed 42; Runs 2 and 3 use seeds 43 and 44.

V. COMPLEXITY ANALYSIS

We conducted a comparative analysis of computational metrics on the WHU-CD dataset under the 5% labeled setting, as reported in Table XII. AMS-SemiCD achieves the highest IoU of 83.13%, while requiring 240.90M parameters, 14.02 min/epoch, and 22.24 GB of training memory. Despite its large parameter count, the model maintains a moderate computational complexity of 53.69 GFLOPs because Mamba performs global context modeling with linear spatial complexity. At 53.69 GFLOPs, AMS-SemiCD requires less computation than SemiCD, RCL, FPA, C2F, and CutMixCD, but more than UniMatch and SemiCD-VL. The 22.24 GB peak memory usage allows the complete framework to be trained on a single NVIDIA RTX 3090 GPU. Compared with UniMatch and SemiCD-VL, AMS-SemiCD improves IoU by 2.33% and 3.89%, respectively, at the cost of a larger model and higher training resource consumption. Overall, AMS-SemiCD prioritizes representation capacity and detection accuracy over model compactness, making it more suitable for accuracy-oriented offline remote-sensing analysis than resource-constrained edge deployment.

TABLE XII
COMPARISON OF PARAMETERS AND COMPUTING COMPLEXITY ON 5% LABELED WHU-CD DATASET

Method	F1	IoU	Params (M)	FLOPs (G)	Train Time (min/ep)	Train Mem (GB)
SemiCD [49]	77.57	63.36	46.85	73.21	7.05	13.37
RCL [50]	80.45	67.30	46.85	73.21	6.95	12.49
FPA [51]	83.52	71.70	46.85	73.21	6.45	15.48
UniMatch [33]	89.38	80.80	40.47	29.10	5.53	14.69
C2F [9]	85.97	75.40	16.26	62.02	5.97	10.56
SemiCD-VL [40]	88.42	79.24	26.13	21.60	5.92	12.23
CutMixCD [35]	87.27	77.42	46.85	73.21	5.40	12.09
AMS-SemiCD	90.79	83.13	240.90	53.69	14.02	22.24

VI. CONCLUSION

In this paper, we introduced AMS-SemiCD, a novel framework for SSCD designed to address the critical challenges of unreliable pseudo-label generation, inefficient unlabeled

data utilization, and inadequate feature extraction for complex bitemporal changes. The framework integrates two synergistic components: ASPR, which employs a dual-adaptive strategy to ensure pseudo-label quality and maximize the utility of unlabeled data, and MSE-Net, a hybrid CNN-Mamba architecture for robust and discriminative feature representation.

Comprehensive experiments conducted on benchmark datasets validate that our proposed method achieves superior performance, demonstrating its effectiveness and robustness. Despite these promising results, we acknowledge certain limitations. First, although Mamba avoids quadratic spatial complexity in global context modeling, the dual-branch architecture introduces a large parameter count and higher training time and memory consumption, limiting deployment in resource-constrained scenarios. Future work will investigate parameter-efficient architectures, model compression, and knowledge distillation to reduce resource requirements. Second, the current evaluation focuses on binary change detection, while the generalization of AMS-SemiCD to multi-class scenarios with greater semantic complexity remains unexplored. Extending the framework to multi-class and multimodal change detection represents an important direction for future research.

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